

De Montfort University

Beng (Hons) Mechanical Engineering – Year 3

Objective: Design & Analyse the Suspension & Steering Components of the Future 2027 DMU FSAE Race Car

Report: Suspension & Steering

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Abstract

The suspension & steering geometry must be designed for the formula student competition. Designing the suspension optimally, requires collaboration with the upright and chassis designer to understand design limits and negotiate final designs. The aims and objectives of this design challenge are to gain experience in a team, while individually designing the geometry of the most complex and significant section of a race car, the suspension. The real-world limits are cost, manufacturing & Formula Student Competition regulations. Suspension types must be researched and analysed to find the optimal suspension system for formula student racing. Using Susprog3d (a mathematical software calculating the optimal suspension node co-ordinates based on variables of a race car), the optimal suspension nodes must be obtained and prepared for use by the future 2027 De Montfort University Formula Student race team. The final susprog3d design and findings will then be compiled and communicated to the Formula Student DMU race team for future use in competition. Where the future team will finalise the geometry according to the future rules and regulations of that time, producing a 3D CREO design, analysing the design using ANSYS and finally manufacturing it for use in the 2027 race.

Introduction

In the UK the Institution of Mechanical Engineers (IMechE) hosts the formula student competition. It is a competition among universities and takes place annually at Silverstone Circuit. University students compete to design and bring forth a race car under the formula student regulations to race against other universities. The competition is split into two different categories: Static Events (business/design) and Dynamic Events (on-track performance). Static Events cover the engineering design, cost/manufacturing and a business plan presentation (mock commercialisation). Dynamic Events cover a standing start acceleration test of 75 meters, a figure-eight timed circuit, a single lap, endurance and fuel efficiency during the race. Suspension and steering are crucial to making an efficient, fast and safe car in this race.

Suspension is complex, requiring precision to ensure every possible movement and angle is ideal for all racing situations. This requires a balance of all points at which the geometry pivots as one change can cause a domino effect on all values. The suspension is arguably the most influential piece of a car, with the ability to ruin it's potential on a racetrack even though the rest of the car is optimal, and the contrary being possible too.

Aims

The aim of this project is:

- To design the optimal suspension geometry using susprog3d.
- Improve the cars performance in the formula student competition compared to last year.

- Prepare all values and information for the future 2027 team to finalise and manufacture the geometry.

Objectives

This project entails the following:

- Research of suspension and steering components.
- Which type of suspension is best for racing.
- What modifications or add-ons can be applied to improve performance.
- How the upright dimensions must be considered when designing the suspension.
- Using Susprog3d to design the suspension components.
- How the components will be attached to the DMU team race car.
- Develop parts list

Sustainability Goals [\[1\]](#)

1. SDG 4 (Quality Education): This project utilises advanced technical software such as CREO to design the components in 3D, Susprog3d to simulate the kinematics of suspension according to key values and structural analysis using ANSYS FEA. This aligns with the goal of increasing technical skills of persons with the aim of being professional engineers.
2. SDG 9 (Industry, Innovation & Infrastructure): By using Finite Element Analysis (FEA) to optimise the component's weight and strength, this project promotes the efficient use of material and sustainable design innovation.
3. SDG 12 (Responsible Consumption and Production): Using 3D simulation software, less real-life physical prototypes will be wasted attempting to find a solution. Therefore, less materials are used in total to fulfil the same goal, and FEA means safety is not compromised.



Figure 1 United Nations Sustainable Development Goals

Deliverables:

- Common suspension systems, their advantages and disadvantages.
- Which ones are used for racing around a formula style track.
- Which one is best for DMU racing team car based on effectiveness.
- Utilising Susprog3d to calculate optimal suspension nodes (points that will connect to car/wheel).
- Planning a project that requires research analysis, design, collaboration & software skills.
- How to prioritise aspects of a car for a specific aim; should it be lightweight or heavy, complex or simple, what values matter? And how to transfer that into a design.

Methodology

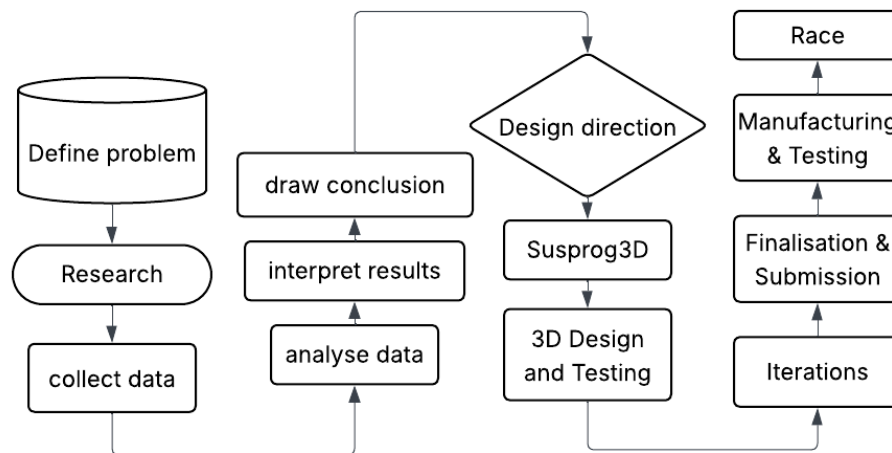


Figure 2 Methodology Steps Flow Chart

Research Gap

- How camber, caster, toe & king-pin inclination can be optimised for racing cars.
- Rules and regulations of formula student race that affects suspension and steering geometry.
- Chassis, Upright, Suspension/Steering dimension limitations.

Research

Title: On the Influence of Suspension Geometry on Steering Feedback [\[2\]](#)

Publisher: MDPI (Applied Sciences)

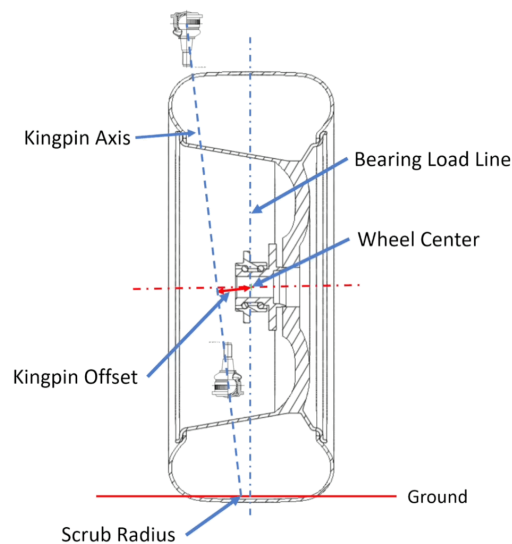


Figure 3 Kingpin Axis, Offset & Scrub Radius Diagram of a Wheel

Kingpin Offset (KO) is critical. Kingpin offset is the primary driver of torque steer and steering disturbance such as bumps or rocks on the road. The KO should be kept as small as possible (ideally near 10-15mm) to ensure the driver receives clean feedback from the tires rather than noise from the drivetrain. This must be negotiated with the upright designer due to possible dimensional limitations.

Scrub Radius (SR) can be lowered to reduce steering effort; however, some SR is necessary for the driver feel the limit of adhesion.

Improper geometry such as excessive KO or SR causes high force peaks when the wheels self-centre. In a formula student endurance event, this leads to driver mental fatigue and inconsistent lap times. The geometry must prioritise self-aligning torque to make the car intuitive to drive.

Title: Design & Analysis of Formula Student Race Car Suspension System [3]

Publisher: ScienceDirect (Procedia Engineering)

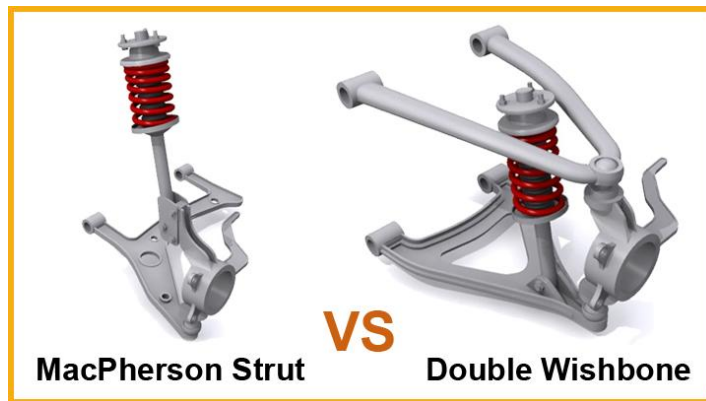


Figure 4 Comparison of Traditional Suspension Against Double Wishbone Suspension Geometry.

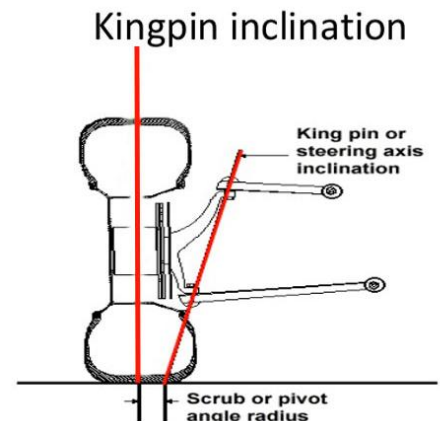


Figure 5 Kingpin Inclination and Scrub Radius

Unequal length non -parallel double wishbone suspension is superior in maintaining the tire contact patch during body roll.

The wishbone pick-up points on the chassis (points that connect to the chassis) must be reinforced to handle braking (longitudinal) and cornering (lateral) loads simultaneously without deflection.

The goal is high roll stiffness with minimal unsprung mass. FEA will be used to justify material choices.

Every gram saved on the uprights and wheels is worth double in performance.

Collaboration with the upright designer is required to ensure Kingpin Axis (the line between the upper and lower ball joints) allows for the desired Caster and Kingpin Inclination (KPI) while leaving room for the brake callipers and hubs.

Title: Race Car Design [4]

Publisher: Derek Seward

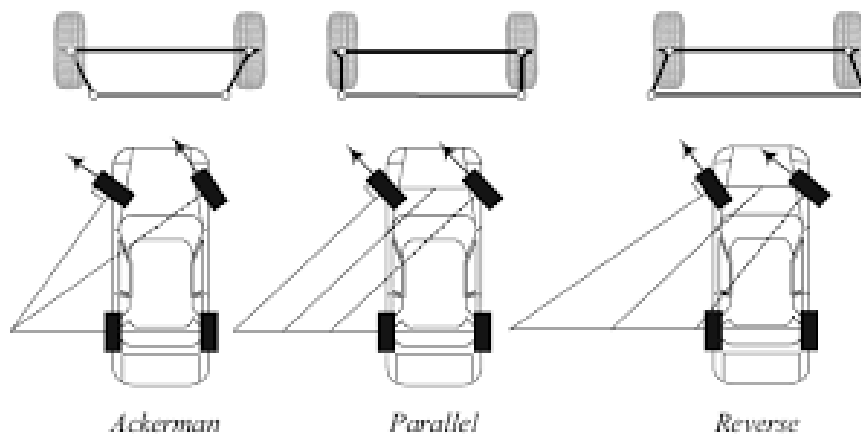


Figure 6 Ackermann vs Anti Ackermann (known as reverse) Geometry

Suspension is a compromise between ride (bumps) and roll (cornering). For FS, where tracks are usually flat but tight, roll control is more important.

Pro-Ackermann geometry optimal for tight hairpins typical of FS tracks. This will require the steering rack to be positioned specifically relative to the front axle line, which must be communicated early to the chassis designer to ensure space for the rack and steering column.

Title: Design Analysis & optimisation of Race Car Suspension [5]

Publisher: IJERT (Internation Journal of Engineering Research & Technology)



Figure 7 Tyre Wear Types from Camber/Caster/Toe & Geometries Diagram

Suspension geometry should be designed to provide negative camber gain during jounce. This counteracts the positive camber caused by body roll, keeping the tire flat on the ground.

The steering tie-rods must be perfectly aligned with the wishbones. Even small amounts like 1mm of error in the steering rack height can cause the wheels to steer themselves as the suspension moves up and down (bump steer), which is a “failure” in FS design judging.

Title: Formula Student Rules & Regulations 2026 [\[6\]](#)

Publisher: IMechE (Formula Student UK)

T4.3 Suspension Requirements: The car must have a functional suspension system with dampers on all four wheels and a minimum travel of 25mm (jounce & rebound)

T4.1 Steering Limits: The steering system must have “positive stops” to prevent the tires from hitting the chassis or suspension members at full lock. There must be zero slack in the steering.

T4.2 Wheelbase/Track: The wheelbase must be at least 1525mm. the smaller of the two tracks (front/rear) must be at least 75% of the larger track.

Rules require a specific tube thickness for the primary structure where suspension attaches. Car must maintain ground clearance of 30mm at any portion of the car except the wheels when a driver is in the car. The rear of the car must have a specific jacking point that doesn’t interfere with the rear suspension links.

Outline Plan

1. Research and communicate findings in report.
2. Decide on suspension type e.g. double wishbone.
3. Decide what components will be used, coil-over, rose joint etc.
4. Use susprog3D to find suspension nodes for such suspension type to be optimal for Formula Student.
5. Iterate and improve until optimal.
6. Prepare co-ordinates for future modelling using Creo
7. Finalise report, communicate findings and final susprog3d design
8. Communicate these findings and the final susprog3d design to the Formula Student team of DMU for future 2027 competition use.
9. Potential improvements from this project

(tab. 1): An initial meeting would be necessary to gain understanding of what is expected in this project, any further meetings with the project supervisor can be planned if necessary.

The focus research is predicted to be short and basic for the general overview of what makes a good suspension geometry.

Meetings with the upright designer (chassis designer N/A), are highly important for use of the suspension geometry. The upright geometry is directly affected by the suspension geometry, as the points designed must connect to the suspension rods. However, there is also a physical limit on how strong the upright can be designed to handle suspension rod connection points if they are too far from the upright, it could mean critical failure is imminent, or more material would have to be added in the design to accommodate for the further our connection points, this would increase weight slowing down the car around a track. So, it is important to find the correct points for the upright co-ordinates of the suspension geometry to ensure ideal dynamics of the car without negatively affecting the upright design.

Furthermore, meetings with the DMU racing team which take place in the university’s workshop, are extremely important. Past designers of suspension geometry can assist by providing recommendations

Finalisation of Design																	
Finalising Writing																	
Submission																	

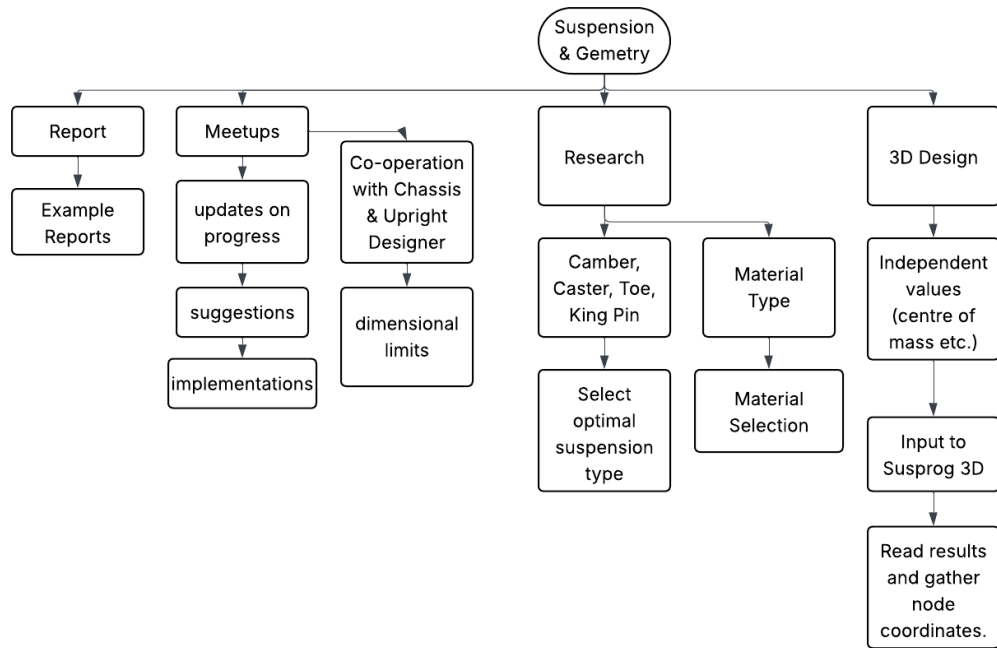


Figure 8 Work Breakdown Structure Flow Chart

Suspension Systems

Suspension Parts

Coil Spring:



Figure 9 Coil Spring

Acts as a simple spring; it is designed to store mechanical energy when a load is applied and release it when the load is removed. They operate based on the physics of elasticity and potential energy, specifically following Hooke's Law, returning to their original shape upon removal of the load.

$$\text{Hookes Law: } F = kx$$

Coil springs support the vehicle's weight, absorb road shocks and help maintain tyre contact with the road for stability. They also offer a more comfortable ride for passengers. However, they release the energy in oscillations, causing a car to bounce after experiencing shock without damping.

Shock Absorber:

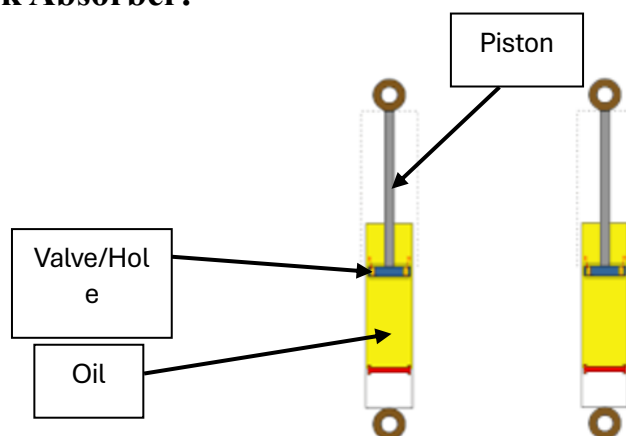


Figure 10 Shock Absorber Animation & Annotation

Functions as a hydraulic oil pump; a piston forces oil through small openings, creating resistance that slows down spring oscillation. When a force is applied on the shock absorber, the piston receives this force. The piston pushes against the oil, and liquids compress a negligible amount, this forces the oil to move through the tiny valve openings. The oil starts to pass the small hole, but due to many factors such as viscous shear stress (key factor), the oil is slowed down and pushes back on the piston. This force back onto the piston slows it down, the viscous stress also converts the initial kinetic energy of the piston into thermal energy.

Coil-Over:



Figure 11 Coil Over Annotated Diagram

coil-over is short for “coil spring over shock” and is the name for a suspension unit that combines a coil spring and a shock absorber into a single integrated assembly. A strut is a specific type of coil-over that also acts as a structural pivot point for steering. In general, “coil-overs” refer to the high-performance, adjustable version of the basic struts or spring/shock setups found in most road cars.

Wheel Alignment

Camber:

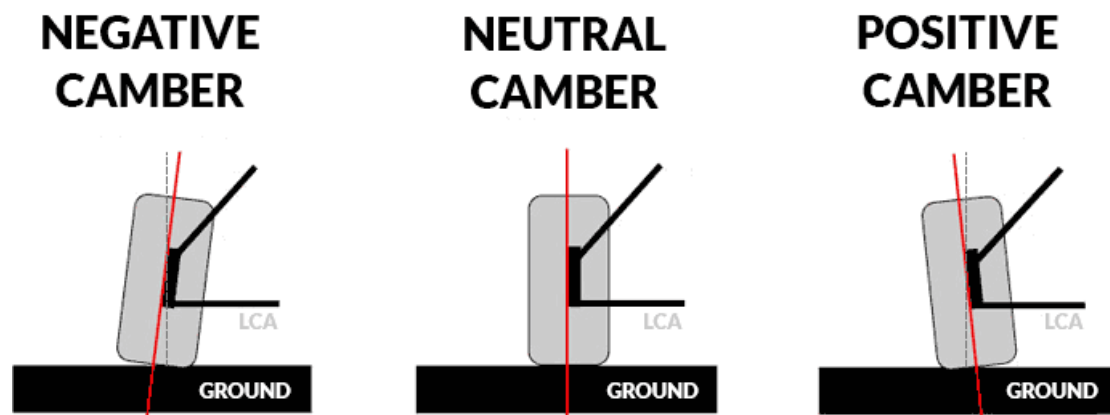


Figure 12 Car Camber Angles Diagram

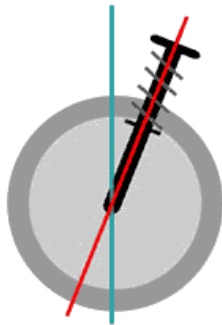
Negative camber is where the top of the wheel is vertically angled towards the car and positive camber is where the top of the wheel is vertically angled away from the car (fig. 3).

Negative camber is widely used for performance driving. when a car turns, it naturally leans (body roll). Negative camber counteracts this, ensuring the tire stays flat against the road during the turn, which provides maximum grip. However, excessive negative camber causes the inner edges of your tires to wear out prematurely and can reduce straight-line braking effectiveness

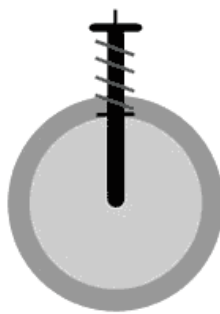
Positive camber is rarely used in modern passenger cars but has roles in heavy-duty and off-road applications, stability under load, easier steering and off-road control. It significantly reduces cornering grip on paved roads and leads to faster wear on the outer edges of the tires.

Caster:

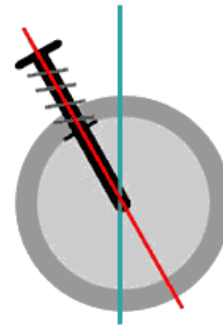
POSITIVE CASTER



NEUTRAL CASTER



NEGATIVE CASTER



← FRONT OF VEHICLE

Figure 13 Car Caster Angles Diagram

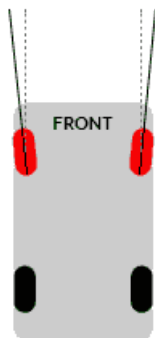
Positive caster is when the steering axis tilts backwards towards the driver, negative caster is when the steering axis tilts forwards away from the driver (fig. 4).

Positive caster is standard for most cars; this geometry makes the wheels naturally want to point straight ahead which prevents the car from “wandering”. It is the reason the steering wheel in cars returns to centre after turning it. Finally, but most importantly for racing, as you turn the wheel, positive caster creates more negative camber on the outside tire, providing extra grip when and where it is needed. This does, however, make the steering heavier.

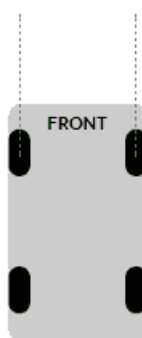
Negative caster is very rare; it makes the wheel light and easy to turn at low speeds. It is very unstable at high speeds.

Toe:

TOE-OUT



NEUTRAL TOE



TOE-IN

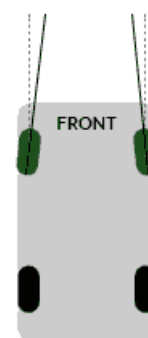


Figure 14 Car Toe Angles Diagram

Toe, often called “tracking”, is the angle of the wheels when viewed from above. Toe-in (positive toe) is where the front edges of the wheels point toward the centre of the car. Toe-out (negative toe) is where the front edges of the wheels point away from the centre.

Toe-in is commonly used in passenger vehicles as it improves straight-line tracking and reduces “wandering” at high speeds. It counteracts oversteer. Excessive toe-in wears the outer edges of the tyres.

Toe-out is often used on the front wheels of track cars to reduce understeer and make the car feel more eager to enter corners. Excessive toe-out causes the inner edges of the tyres to wear down. The main drawback is the car feels more unstable in a straight line, requiring constant steering corrections.

Suspension Classification

Independent:

Allows each wheel to move vertically without affecting the opposite wheel; improves comfort and handling because wheels can independently act against shocks and offer better contact with the road.

Dependent/Rigid:

Connects both wheels on an axle with a solid beam; better for load carrying because they use fewer, but stronger components that offer constant ground clearance with reduced body roll compared to independent systems.

Active/Adaptive Suspension:

Uses sensors and electromagnetic or valve-actuated dampers to change stiffness in real-time; instant optimisation and near-zero body roll, eliminates brake dive/squat among other techniques to improve handling and comfort.

General Terminology:

Sprung/Unsprung mass:

The parts of a vehicle that is supported by the suspension is called sprung weight. Unsprung weight is not supported by the suspension such as the wheels, tires, brakes, axle components and the suspension itself.

Torque Steer:

When front-wheel-drive (FWD) cars pull at the steering wheel under heavy acceleration due to the high-power performance of the car.

Types of Suspension Systems

This section will briefly describe all the present and past conventional suspension setups to provide key understanding. This section will assist in determining the best suited suspension system for the FS competition.

Leaf Springs:

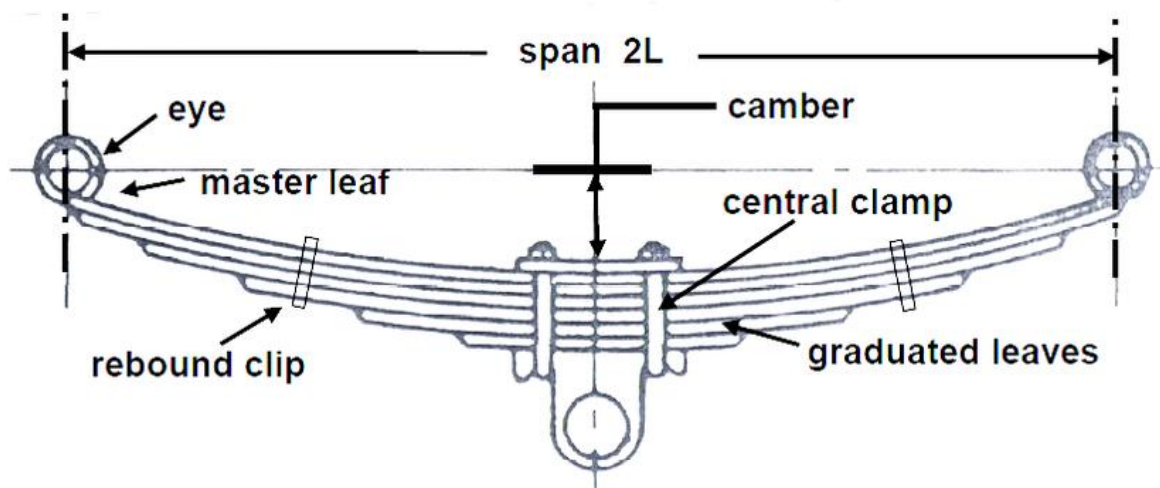


Figure 15 Leaf Spring System

Considered oldest modern type of suspension system, appearing in their metal form on carriages like the Landau around 1750. In 1804, British inventor Obadiah Elliot patented the first truly workable spring-suspension vehicle, using two durable steel leaf springs on each side to attach the body directly to the axles.

Advantages	Disadvantages
• Excel in heavy-duty load-carrying • Towing stability • Cost-effective • Simplistic and reliable	• Stiffer ride • Limited flexibility • Limited adjustability

MacPherson struts:



Figure 7 MacPherson Strut Turning & Load

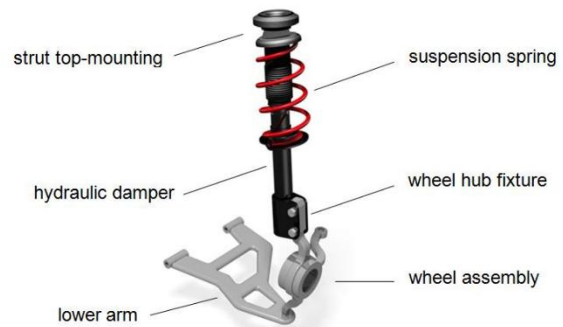


Figure 8 MacPherson Strut Components Diagram

The MacPherson Strut connects the car's wheel directly to the body. It acts as independent suspension allowing for a comfortable and better performing ride compared to other suspension. It houses a coil spring to absorb bumps and a shock absorber inside to stop the car from rebounding.

Table 2 Common Advantages & Disadvantages of the MacPherson Strut

Advantages	Disadvantages
• no need for an upper control arm, compact. • Simple mechanism requires few parts. • cost-effective. • It is lightweight and strong due to its simplicity and geometry. • Low unsprung mass	• No camber gain • Limited adjustability • It can transmit more road noise and vibration into the cabin compared to more complex systems. • Top strut mounts can wear out over time.

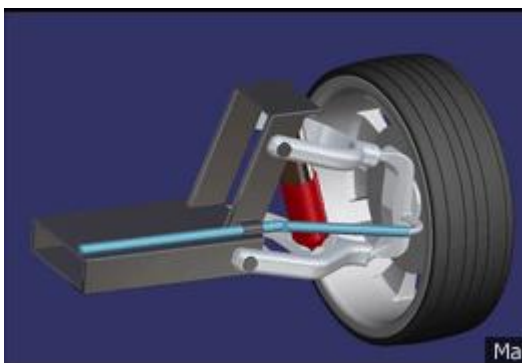


Figure 17 Double Wishbone Suspension Dynamics Animation

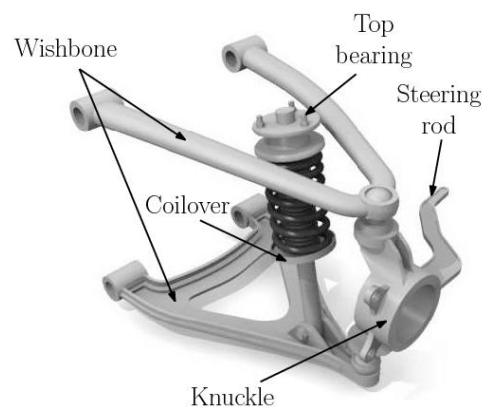


Figure 16 Double Wishbone Annotated Diagram

Double Wishbones:

As shown in (fig. 18), a double wishbone consists of two A-shaped control arms that point towards the wheel, a coil-over (coil spring and damper system) and a top bearing to allow for smooth rotation of the wheel. Commonly, the upper wishbone is shorter than the lower wishbone, this creates negative camber gain.

Table 3 Advantages & Disadvantages of the Double Wishbone Suspension

Advantages	Disadvantages
• It gains negative camber under load; improved handling • Highly adjustable especially when using push/pull-rod system • High overall race performance	• It is much larger than the MacPherson strut. • It requires more parts, and more complex forgings which makes it more expensive to manufacture and service. • More components generally mean more “unsprung weight”.

Double wishbone suspension is used in F1 and supercars for their optimal tyre contact patch at all times. Luxury saloons for a smooth ride without sacrificing handling. Off roaders for their strength under massive vertical travel. Compared to the MacPherson strut, it provides better strength and grip around corners but sacrifices simplicity, cost and size.

Multi-Link:

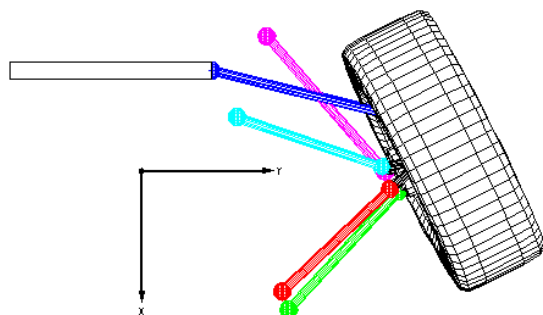


Figure 19 Adjustment Effects of a Multi-Link Suspension System

Multi-Coloured Arms

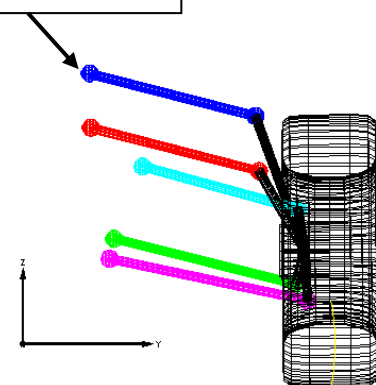


Figure 18 Annotated Multi-Link Suspension Diagram

A multi-link suspension system uses a similar concept to the double-wishbone suspension but, instead of two A-shaped arms, the geometry of the arms is split into separate and independent arms (fig. 9). Each of these arms is a rod with its own bushings at each end. Engineers can change the angle, length and mounting point of each rod to control exactly how the wheel moves in 3D space (fig. 10). These rods are supported generally using a coil-over system in between the top and bottom rods, like a double-wishbone setup, to provide support for the rods under the weight of the car, damping and maintain grip under road shocks. Since the suspension points are individual and adjustable, camber, toe-in/out can all be controlled. This gives the potential for a smoother ride and perfect contact patch in more situations.

Table 4 Advantages & Disadvantages of the Multi-Link Suspension

Advantages	Disadvantages
• Ride comfort • Compact • High grip performance though worse than double wishbone	• Complex to design • Cost, more parts, complexity. • Maintenance, complexity.

Torsion Beams:

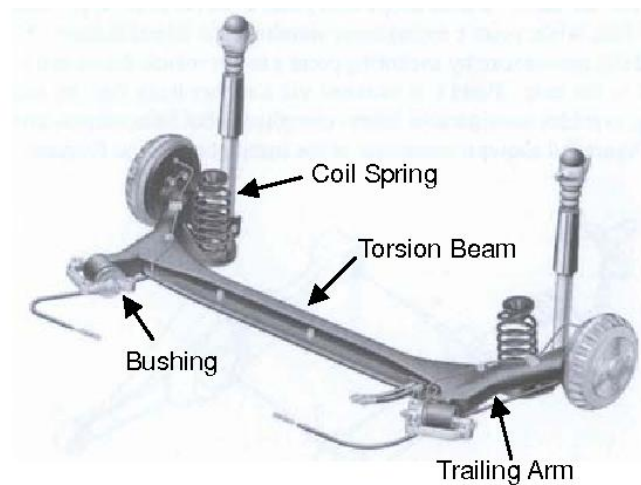


Figure 1. Typical torsion beam rear suspension [2].

Figure 20 Annotated Diagram of a Common Traditional Torsion Beam Suspension Setup

Torsion beam suspension is a type of rear suspensions used primarily used in front-wheel-drive cars. It acts as a mixture of independent and dependent suspension at the same time. It is not independent, but it is flexible enough to act as such under most situations especially for people using a car with torsion beam suspension for general commuting.

Table 5 Advantages & Disadvantages of the Torsion Beam Suspension System

Advantages	Disadvantages
• Cost: cheap to manufacture and easy to install. • Space: Compact; allows for a flat trunk floor and larger gas tank. • Durability: fewer moving parts means less to wear out or break. • Alignment: wheel alignment stays consistent as the body leans.	• Limited tunability for high-performance. Cannot adjust camber or toe easily. • Not truly independent; one wheel affects the other. • Bumps mid-turn can cause momentary instability and a feeling of “hedgehopping”.

Advanced Air Suspension:

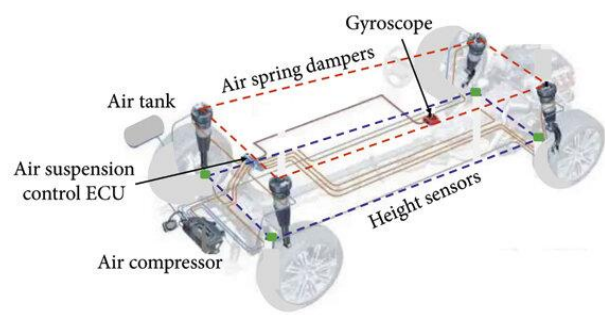


Figure 21 Air Suspension System Annotated Diagram

Air suspension systems replace traditional suspension systems and can be utilised alongside advanced sensors and an ECUs coded by engineers. This system is called advanced air suspension and can provide perfect riding comfortability. Ride height sensors monitor distance.

Table 6 Advantages & Disadvantages of the advanced air suspension system

Advantages	Disadvantages
• Ride Quality: ride height and suspension stiffness can be instantly adjusted for future road shocks and added weight, keeping the car completely flat. • Clearance: ride height can be adjusted to get over bumps without bottoming out. • Aerodynamics: car can drop slightly at high speeds to improve fuel economy.	• Cost: high initial cost • Complexity: Difficult to code and engineer, maintenance is a nightmare, issues can lead to the car being inoperable until fixed. • Extremely heavy, compressor, air bags etc.

Trailing Arm Suspension:

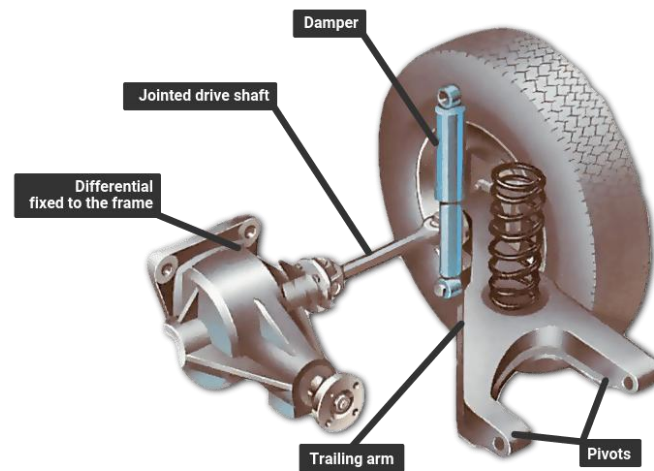


Figure 22 Trailing Arm Suspension Example Annotated Diagram

Is an independent suspension system. Closely related to the torsion beam system; doesn't have a connecting beam from one wheel to another. Instead, there are two trailing arms for each wheel, that connect to the chassis at a pivot point and allow for vertical movement independent of the other wheel (fig. 16). A coil-over is placed on the arm and the point at which it is mounted significantly impacts how the suspension acts: closer to the wheel the stiffer it acts.

Advantages	Disadvantages
• Rugged: strong geometry and compact size • Cost: few parts, cheaper to manufacture and faster to assemble. • Predictable: stable for high-speed cruising.	• Contact Patch: pure trailing arm setups act as a rigid component when it comes to camber • Due to the pivoting movement, the wheel moves forward and back when suspension is activated, causes instability on very bumpy roads.

Chosen suspension

The double wishbone suspension. Specifically, the pull-rod connected to the bottom a-arm system. Firstly, the double wishbone suspension system provides ample adjustments, allowing the tuning of; camber, caster, toe, kingpin, Ackermann etc. Not to mention the ability to adjust how these values change under bump/droop.

Bump/droop is when the suspension compresses and elongates (wheel travelling up and down compared to static).

A push-rod system is where an additional arm connects from either the upper or lower a-arm of the double wishbone to a coil-over and finally to the chassis. It acts directly according to the suspension, compressing when the suspension compresses and vice versa.

The pull-rod system is an alternative to the push-rod system where a rod is most commonly connected to the top a-arm, from there, downwards, to the chassis where it will connect to a coil-over. The difference is this allows for a lower centre of gravity, which is advantageous, but the coil-over being placed on the bottom of the car causes issues. It is much harder to accommodate a coil-over at the bottom of the FS car since the car is low to the ground and there isn't much space in-board (inside chassis) for anything other than the driver and pedals at the front.

A push-rod system is easier to maintain and adjust since it is more accessible. It also provides more space for a wider range of angles and therefore adjustability.

Formula Student Competition

Designing a good suspension would be futile if the FS car would not qualify for the race. This section provides the relevant rules and regulations that must be taken note of before designing a FS race car's suspension & steering component.

Key Rules & Regulations

1. **Suspension Rule T 2.5.1** “the vehicle must be equipped with fully operational front and rear suspension systems including shock absorbers and a suable wheel travel of at least 50 mm and a minimum jounce of 25 mm with driver seated”.
2. **Ground Clearance Rule T 2.2.1** “The minimum static ground clearance of any portion of the vehicle, other than the tires, including the driver, must be 30mm”.
3. **Wheelbase Rule T 2.9.1** “The vehicle must have a wheelbase of at least 1525mm”.
4. **Track Width Rule T 2.9.2** “The smaller track of the vehicle (front or rear) must be no less than 75% of the larger track.”
5. **Steering System Rule T 2.8.2** “The steering system must have positive steering stops that prevent the steering linkages from locking up. The stops must be placed on the rack and must prevent the tires and rims from contacting any other parts”.
6. **Steering System Materials Rule T 2.8.4** “Allowable steering system free play is limited to a total of 7° measured at the steering wheel”.
7. **Wheel Clearance Rule T 2.6.4** “The radial clearance between any non-rotating part and the inside of the rim must be at least 5mm in static condition at any steering angle and any ride height”.
8. **Suspension Visibility Rule T 2.5.2** “All suspension mounting points must be visible at Technical Inspection, either by direct view or by removing any covers”.

Dynamic Events

1. **Skid Pad – 75 points** A figure-of-eight course. Inner circles are 15.25m in diameter and outer circles 21.25m, with the centres 18.25m apart. Driving path is 3m wide.
2. **Acceleration – 75 points** A straight line sprint of exactly 75m from standing start to finish line, at least 3m wide with cones at 5m intervals.
3. **Autocross / sprint – 100 points** A handling track under 1.5km in length built to these guidelines; **rule D6.1.1**, “straights: no longer than 80m, constant turns: up to 50m in diameter, hairpin turns: minimum of 9m outside diameter, slaloms: cones in straight line with 7.5m to 12m spacing”
4. **Endurance and Efficiency – 250 points + 100 points** these two are scored separately but run as one event. Rules **D7.1.2** and **D7.1.3** “the lap length is approximately 1km and the total distance is approximately 22km. There is a mandatory driver change at the midpoint (after 11km). The efficiency score is calculated from fuel or energy consumption during the endurance run”.

Since the race is primarily based around narrow, sharp turns, positive Ackermann is optimal for the FS competition.

Suspension Components

Coil Over

(fig. 23): this technical drawing provides values that must be entered in susprog3d for accurate results

During the process of designing the geometry using susprog3d, when using the lowest provided spring rate – the least stiff spring - the calculated Hz of the spring was higher than the target before adjustments using the bell crank.

DBB2.25/250:

- Inner diameter: 36.3mm
- Spring rate: 250lb/inch
- Free length: 122mm

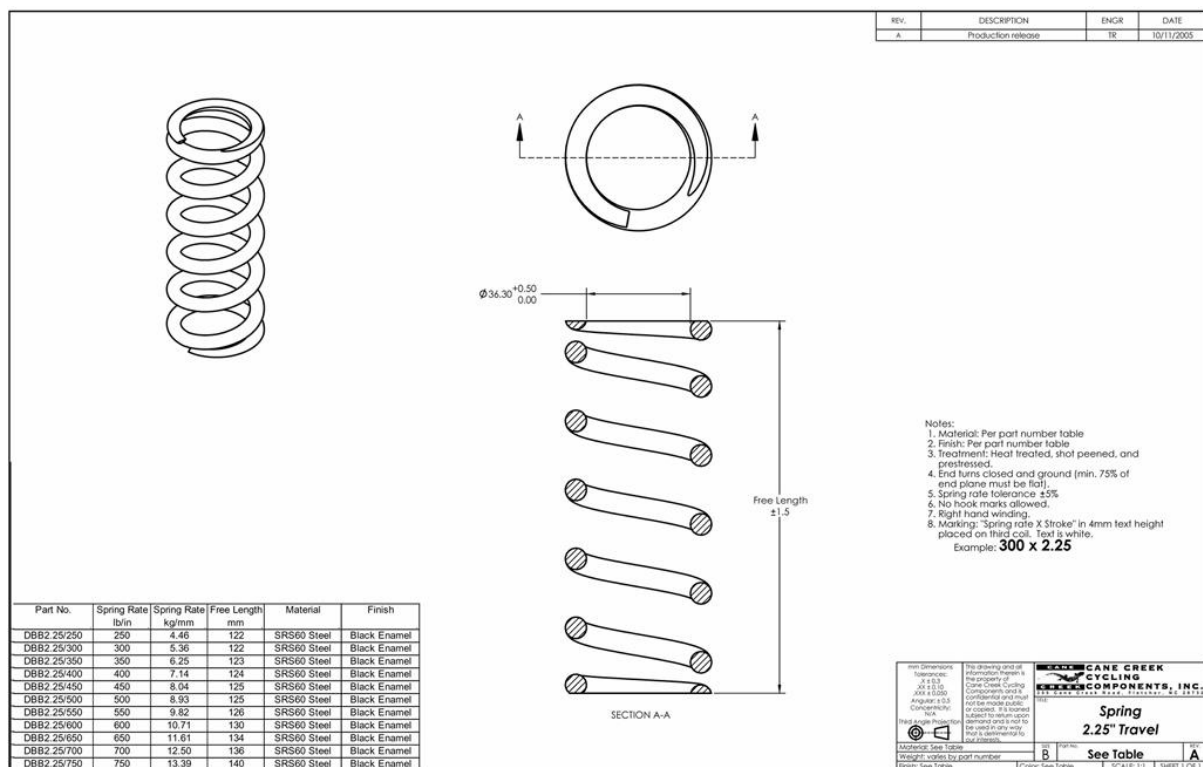


Figure 23 Ohlin's FS coil over spring dimensions & spring rate options

(fig. 24): clearance for the coil-over housing should be considered when designing suspension geometry. More importantly the length between the mount centres determines the mounting geometry in susprog3d, allowing for preload adjustment and proper mounting without the use of extensions – which add weight.

- Length between mount centres: 200mm

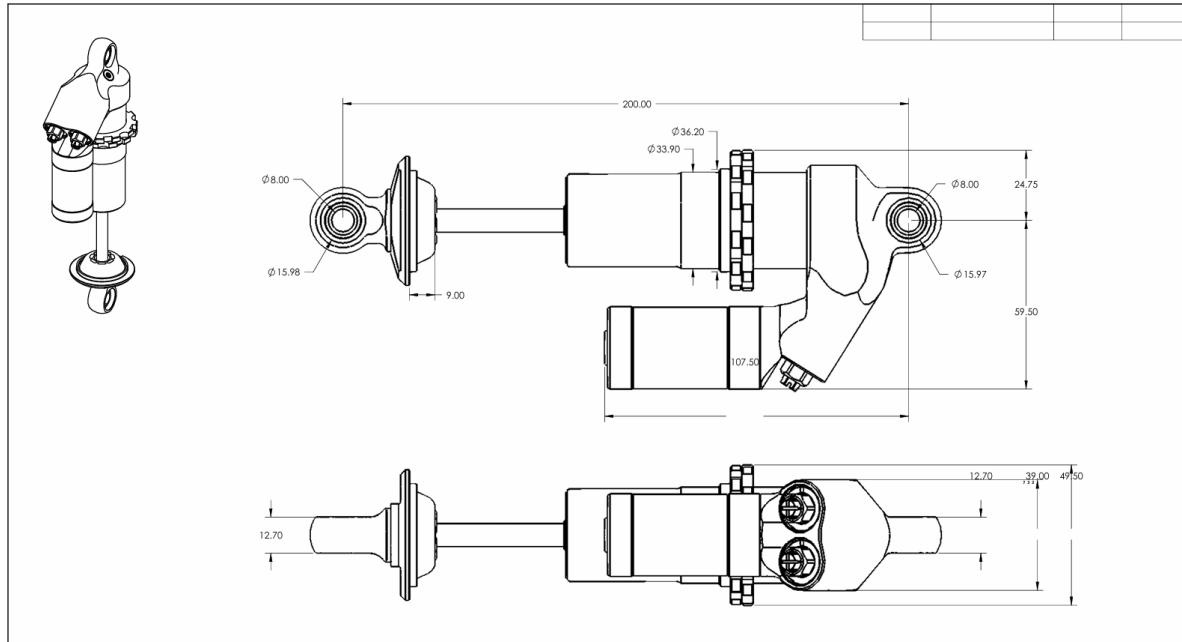


Figure 24 external schematic of the Ohlin's FS coil-over damper system

(fig. 25) another important value is how far the coil-over can compress – the stroke – which according to the Ohlins website is 57mm.

[Link](#)

SPECIFICATIONS:

- Overall length = 200mm (center to center of spherical bearings, fully extended)
- Stroke = 57mm
- Weight = 394g without spring
- Spherical Bearing dimensions:
- ID = 8 mm
- Ball Width = 8 mm
- OD = 15 mm

COMPATIBLE SPRINGS:

Please refer to the 'Spring Details' document in the Technical Information section of this page for general dimensions of compatible springs that fit within the constraints of the damper.

Hyperco-branded springs for the TTX25 are readily available through several of our dealers, including Essex Racing and KAZ Technologies.

If a yellow Ohlins spring is preferred, our 18075-series mountain bike springs are a great option. However, they are only available in a limited range of spring rates. These springs can be purchased directly from our website or by contacting your local Ohlins dealer.

REBUILD KIT:

Figure 25 Ohlin's website with key information on the FS coil over

(Fig. 26): price for coil over at 1004.30 euros or £869.52 each. Four for the whole car would cost £3478.08.

[Link](#)

PROFESSIONAL MOTORSPORT TTX25 MKII FSAE FORMULA STUDENT 200/57 MM

Öhlins has been present and dominated the racing scene for many years. It's been the ground where our products can be tested and no short cuts are taken to find the last bit of performance. Öhlins cover the whole racing scene, from Formula One and single seaters to prototypes, touring cars, GT Racing and stock cars. From specific shock absorbers to our kit system, there isn't a racing car where Öhlins shock absorbers can't fit.

The TTX 25 MkII FSAE uses our well-proven twin tube technology, in a small package. The 4-way externally adjustable damper is completely rebuildable and offers the possibility for setting changes. With a wide range dynamometer plots, CAD models, and other technical information, the TTX25 FSAE provides the perfect package for Formula Student racing teams.

PRICE AND AVAILABILITY

1.004,30 €

830,00 € plus 21,0% VAT (Belgium)

in stock

(3 - 5 business days)

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 Datasheet

The picture shown can deviate from the product delivered.

Figure 26 Öhlins TTX25 MKII FSAE FS coil over price

Rose Joints

Rose joints are used for the mounting of the suspension and steering arms to the upright and chassis. They allow for some angular movement which is perfect for a FS race car suspension.

An alternative would be a ball joint. However, these have higher levels of compliance, fixed geometry – rose joints allow the suspension arms to be extended to change camber, caster, toe... ball joints are heavier and are primarily designed for vertical load bearing.

(fig. 27): This rose joint has been used for the last DMU FS race car. It has a larger thread size than bore size to withstand higher forces applied on the thread, while allowing for smaller mounting bolt which saves weight. UNF standard, provides a finer thread than coarse thread, which resists vibrations and provides higher strength, these are important characteristics for load bearing and moving / vibrating components such as car suspension.

[Link](#)

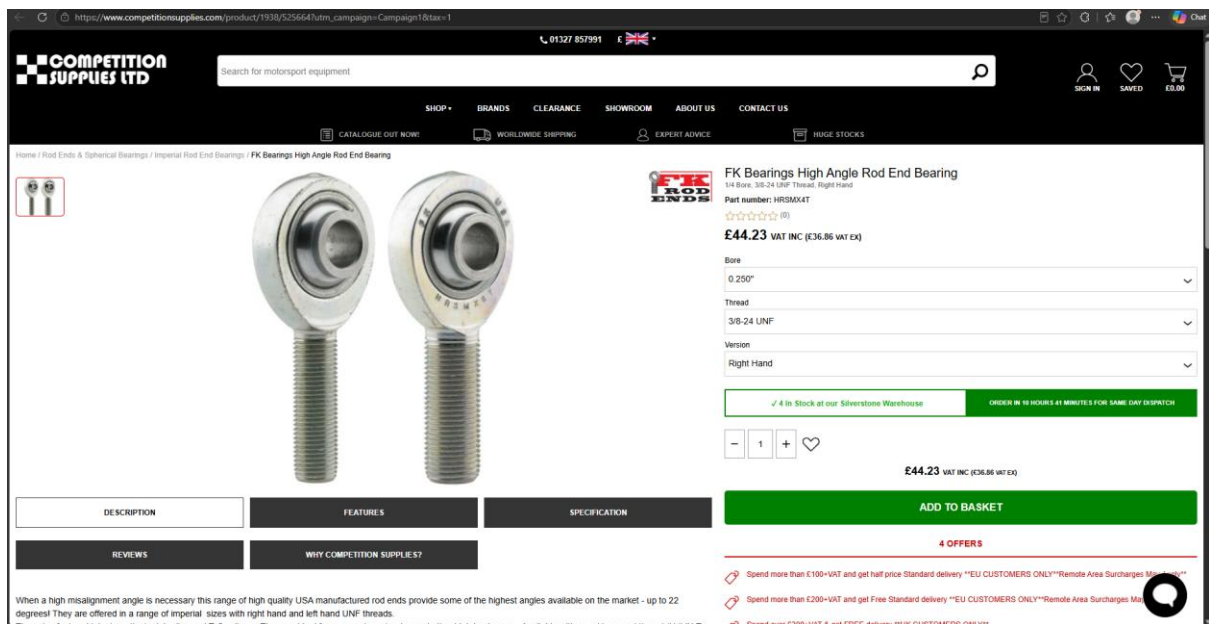
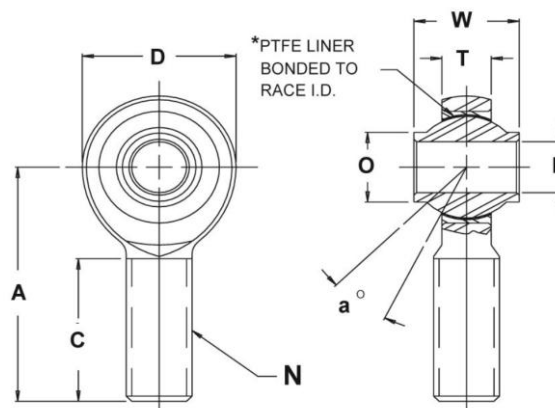


Figure 27 3/8" unf thread x 1/4" bore, rose joint

(fig. 28): the max load capacity is 10,790 lbs or 4894kg. This is much more than the FS car's weight or any dynamic load it may face during the competition.



HRSMX-T High Misalignment Male Rod End - Heavy Duty									
B Dia	W Width	Ball Dia	A Lgth	Thread UNF	C Lgth	Max Load (Lbs)	A° Mis Angle	Part No L/H	Part No R/H
0.250"	0.593"	0.593"	1.938"	3/8-24	1.187"	10,790	23	HRSMXL4T	HRSMX4T
0.3125"	0.813"	0.781"	2.125"	7/16-20	1.281"	11,789	22	HRSMXL5T	HRSMX5T
0.375"	0.813"	0.781"	2.125"	7/16-20	1.281"	11,789	22	HRSMXL6T	HRSMX6T
0.4375"	0.875"	0.875"	2.438"	1/2-20	1.468"	17,100	21	HRSMXL7T	HRSMX7T
0.500"	0.937"	1.000"	2.625"	5/8-18	1.562"	23,703	19	HRSMXL8T	HRSMX8T
0.625"	1.200"	1.250"	2.875"	3/4-16	1.687"	32,100	19	HRSMXL10T	HRSMX10T
0.750"	1.280"	1.375"	3.375"	7/8-14	2.000"	38,701	18	HRSMXL12T	HRSMX12T

Figure 28 rose joint engineering drawing & specification, highlighted in red is the selection being made

Bolt / Nut for Suspension Arms

(fig. 29): This bolt is military grade certified, meaning the chemistry must follow AISI 8740 (Nickel-Chromium-Molybdenum steel), AISI 4140 (Chromium-Molybdenum steel) or AISI 4037 (Molybdenum steel). A strict hardness of 26 to 32 Rockwell C – sweet spot for some elasticity to bend slightly rather than shattering in event of a collision - and a minimum tensile yield of 851 MPa.

Rule T10.1 suspension bolts are explicitly classified as “critical fasteners”. The rules state they must be a minimum of metric grade 8.8 or SAE grade 5. These AN (army / navy) bolts are roughly equivalent to 10.9. Making them a suitable choice.

These bolts are extremely cheap at only 24p each.

[Link](#)

The screenshot shows the Skycraft website's product page for a military grade bolt. The header features the Skycraft logo, contact information, and a navigation menu. The product is identified as 'Bolt 1/4D. 15/16G. 1-13/32L.' with a part number 'AN4-13A'. The price is listed as £0.24 (including VAT), and it is noted that 76 units are in stock. A quantity selector is set to 1, and there are buttons for 'Add to cart' and 'Buy with shop'. A note mentions that pickup is available at Riverside House, Holbeach, and is usually ready in 4 hours.

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Home ▸ Hardware - Imperial ▸ Bolts ▸ AN4 Bolts ▸ Bolt 1/4D. 15/16G. 1-13/32L.

Bolt 1/4D. 15/16G. 1-13/32L.

P/No: AN4-13A

£0.24 (£0.29 including VAT)

Shipping calculated at checkout.

76 IN STOCK

Quantity

1

Add to cart

Buy with shop

[More payment options](#)

✓ Pickup available at Riverside House, Holbeach
Usually ready in 4 hours
[View extra information](#)

Figure 29 military grade bolt, 1/4-inch diameter, 0.9375" grip length, 1.406" total length

(fig. 30): under that part number “MS21042”, the design specifications fit that of a 1/4-inch self-locking, military standard nut. Both the bolt and this nut are designed to withstand heat and corrosion making them a perfect choice for a FS race car.

This part is meant to be used alongside the bolt in (fig. 29) and the nut (fig. 31) on the a-arms, steering arm and pushrod upright mount.

[Link](#)

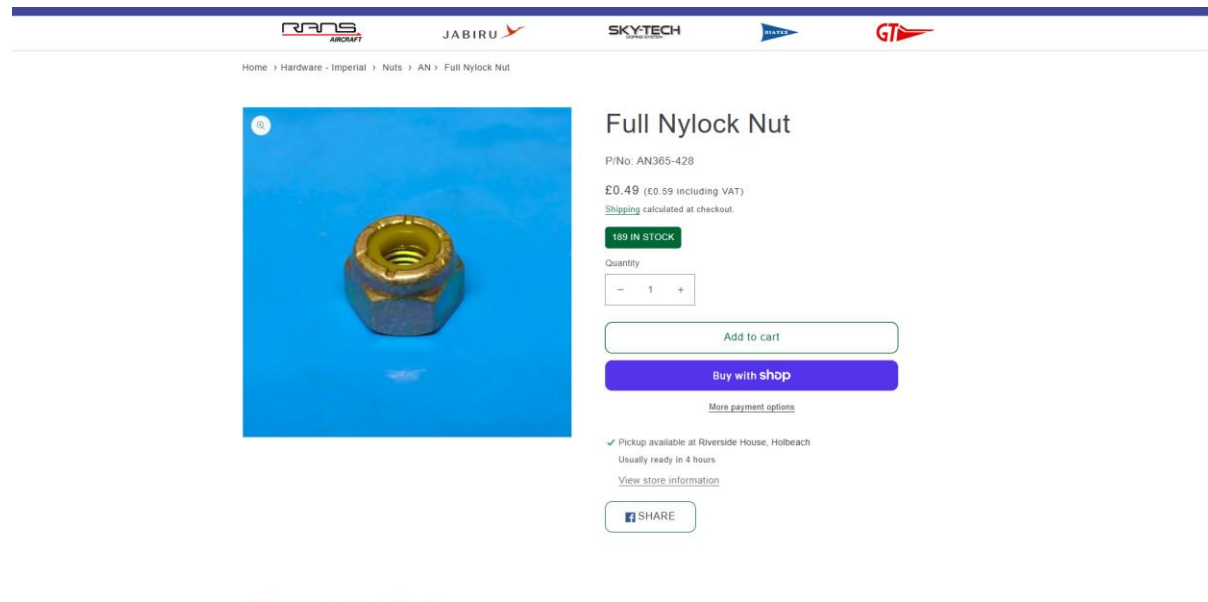


Figure 30 military grade, 1/4-inch nut

[Link](#)

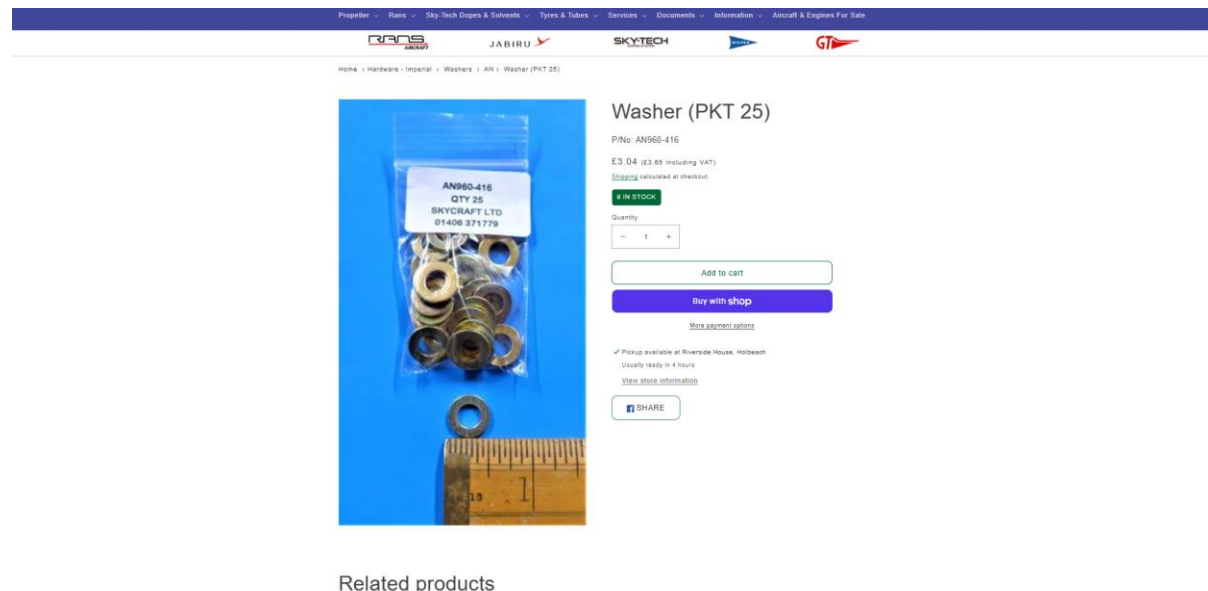


Figure 31 military standard washers for the 1/4-inch nut and bolt

Bolt / Nut for Pushrod and Coil Over System

(fig. 32): since the ohlins coil over has a mount hole of 8mm diameter, M8 bolts, nuts and washers must be used. Grade 12.9 is strong enough to handle suspension dynamic spikes. 160mm to accommodate for mounting brackets, spacers can be used to make up for excess.

[Link](#)

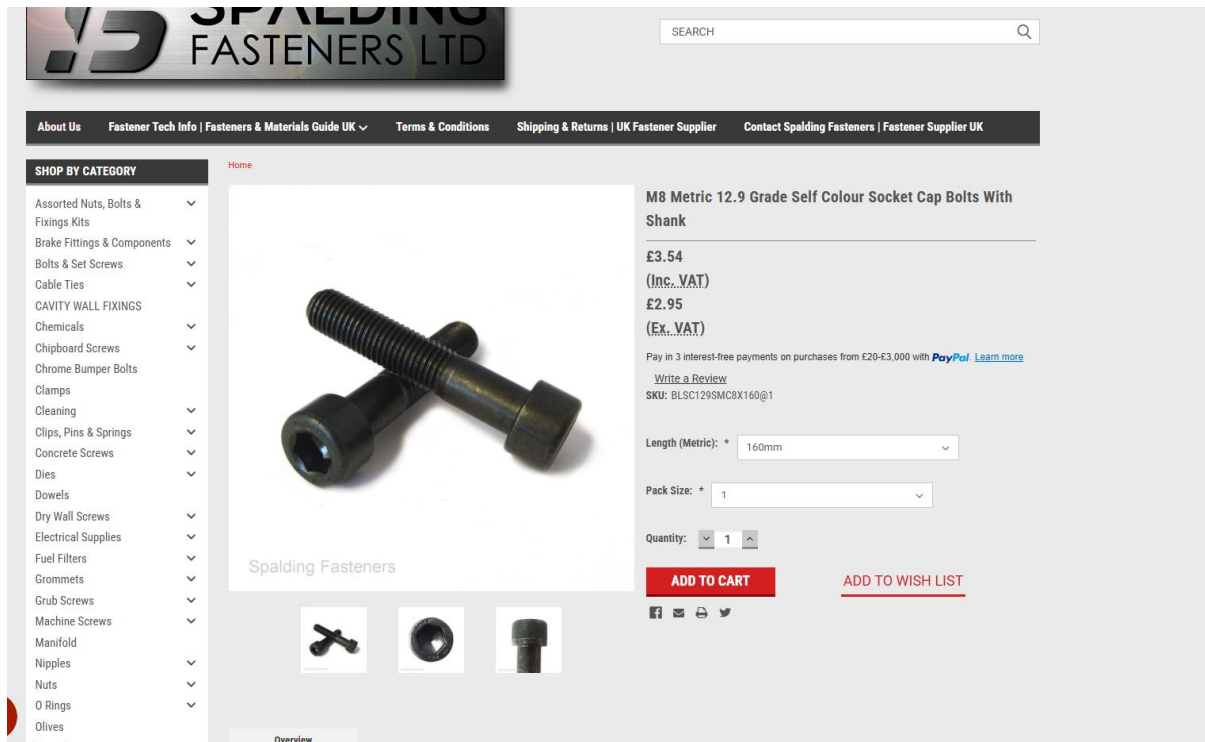


Figure 32 grade 12.9, 160mm length, M8 bolt

(fig. 33): certified under ISO 898-2, property class 10/12. The bolt in (fig. 32) and the washer (fig.34) are to be used together with this locknut to mount the pushrod onto the ohlins coil over (fig. 25), and on the coil over to the chassis mount.

[Link](#)

Lock Nut M8 X 1.25 Right Hand Thread

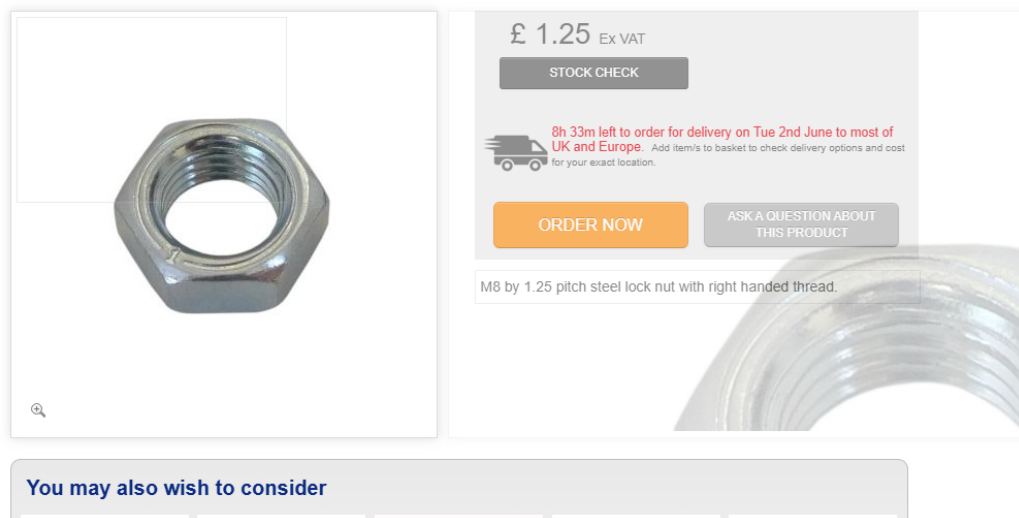
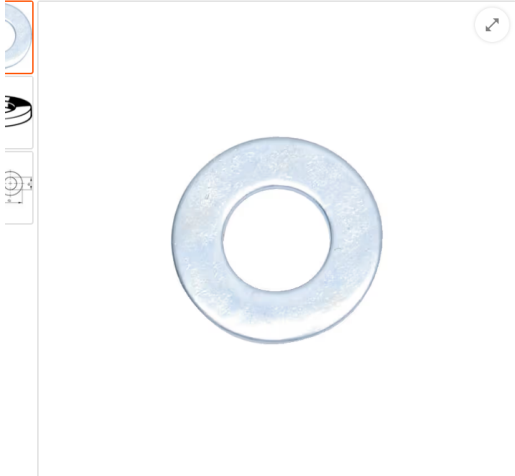


Figure 33 M8 x 1.25 lock nut

(fig. 34): certified to ISO 7089, hardness class 300 HV.

[Link](#)



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Product Specification

Reviews

Figure 34 M8 washer, hardened steel

Susprog3d

How it works

Susprog3d is a software which works using mathematical calculations to present results as values for important dynamics of the car. The software uses values such as the unsprung mass of each corner, or the co-ordinates for each point of the double wishbone geometry to calculate these values.

Below is an example:

(fig. 35): the suspension geometry is presented in 3d from a plan view where the red arms are towards the front of the car. A text box is displayed with captions and the choice of value. These values are co-ordinates for the four a-arm joint mount points, and the steering arm mount point. These values determine where the inner ball points will be in this plane according to an origin they all share. The chassis joints are the points that connect to the chassis and not the upright. The upright is within the wheel's rim.

The points which the co-ordinates control, are highlighted with circle markers. The grey point marks a hidden ball joint.

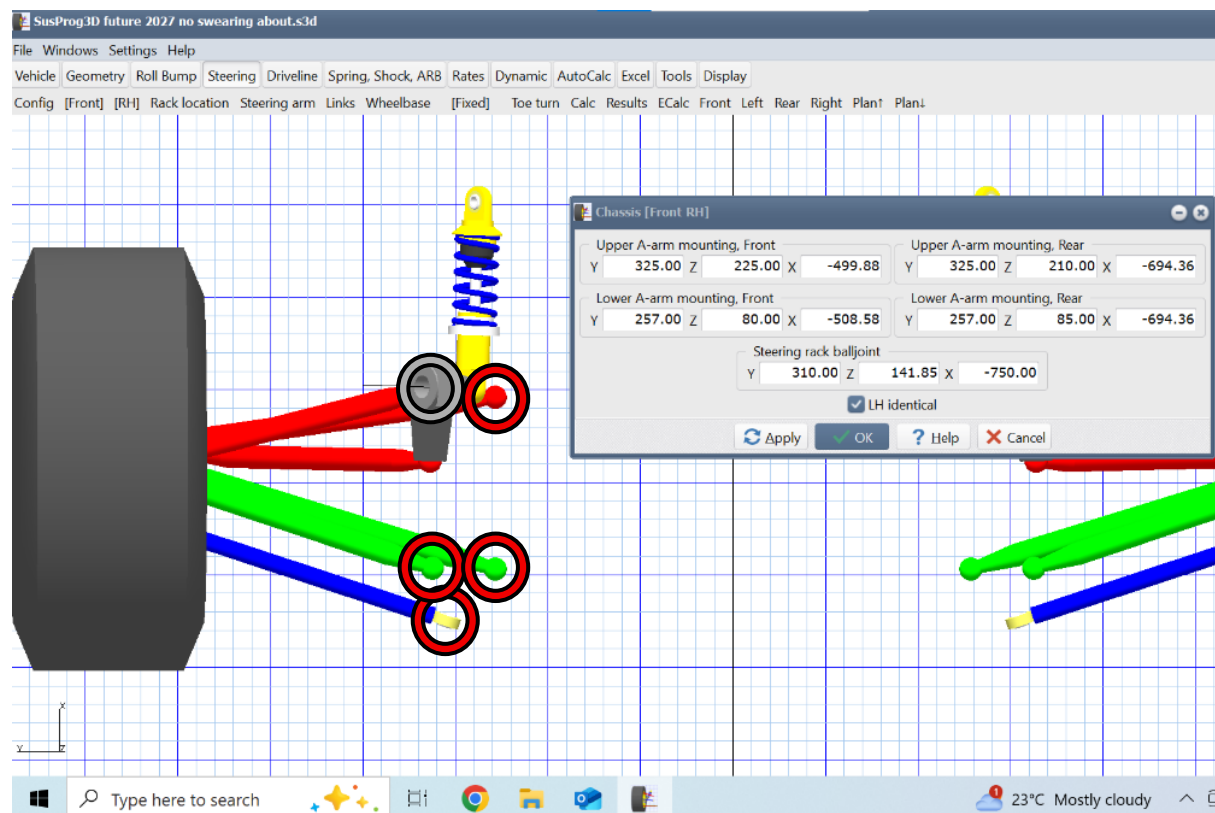


Figure 35 front suspension, plan view, chassis arm points

(fig. 36): the same 3d geometry is presented but in a left side view, where once again the red arms signify the side towards the front of the car. Another text box is displayed this time with only three co-ordinates and some other selections. The “upright type” selection gives a choice of steering geometry type, the upper and lower ball joint means the software will determine the caster, kingpin inclination and scrub radius through a virtual steering axis from the upper and lower arm ball joints. The co-ordinates control only three points because double wishbones connect to the same point, meaning both upper arms connect to one point and so do the lower; they both connect to the upright. The third point is for where the steering rod connects the upright. These co-ordinates have an origin of the centre point of where the upright would connect to the wheel.

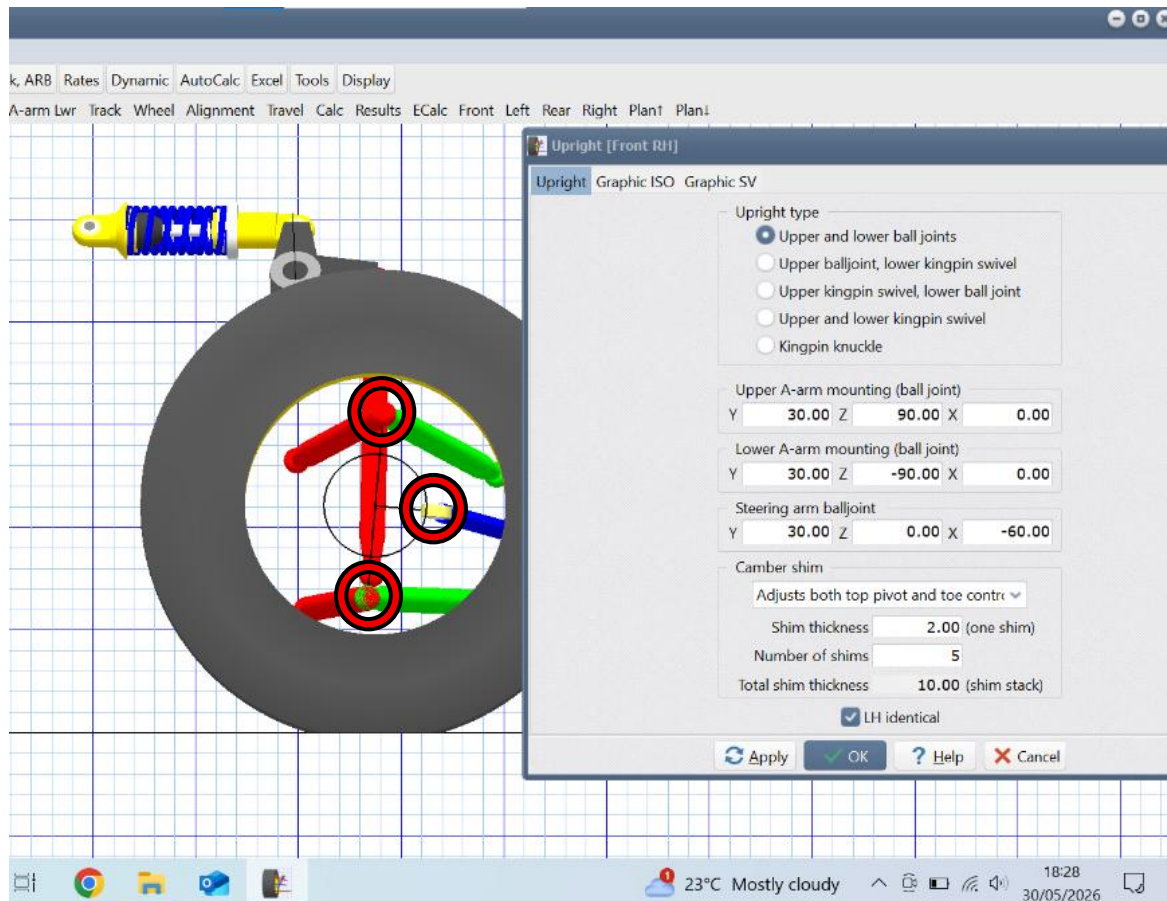


Figure 36 front suspension, left view, upright arm points

Fig. 37 shows some important values. The highlighted values, roll centre height and brake anti-dive represent the peak of key values in assuring ideal dynamics. These change according to the previously mentioned arm co-ordinates.

Roll centre height is the imaginary point at which the car rolls around, valued in mm from the ground at which the tyres rest on. The higher the roll centre the less body roll there is, but the higher the jacking (lift) force there is on the vehicle's chassis. And vice versa for a lower roll centre. A good balance would mean the vehicle experiences somebody roll to give the driver feedback, low jacking force to maintain grip and a setup where the roll centre moves very little as the car dives, squats, and rolls. So, this 50mm value isn't complete without knowing how it changes under different dynamics, which can be shown in another menu.

The next value is the brake anti-dive. A team member recommended the brake anti-dive to be increased from last year's car, which was at 25%. The new 31.2% confirms an increased value.

Using susprog3d is learning what the target should be, assigning target values and changing the co-ordinates until these targets are satisfied.

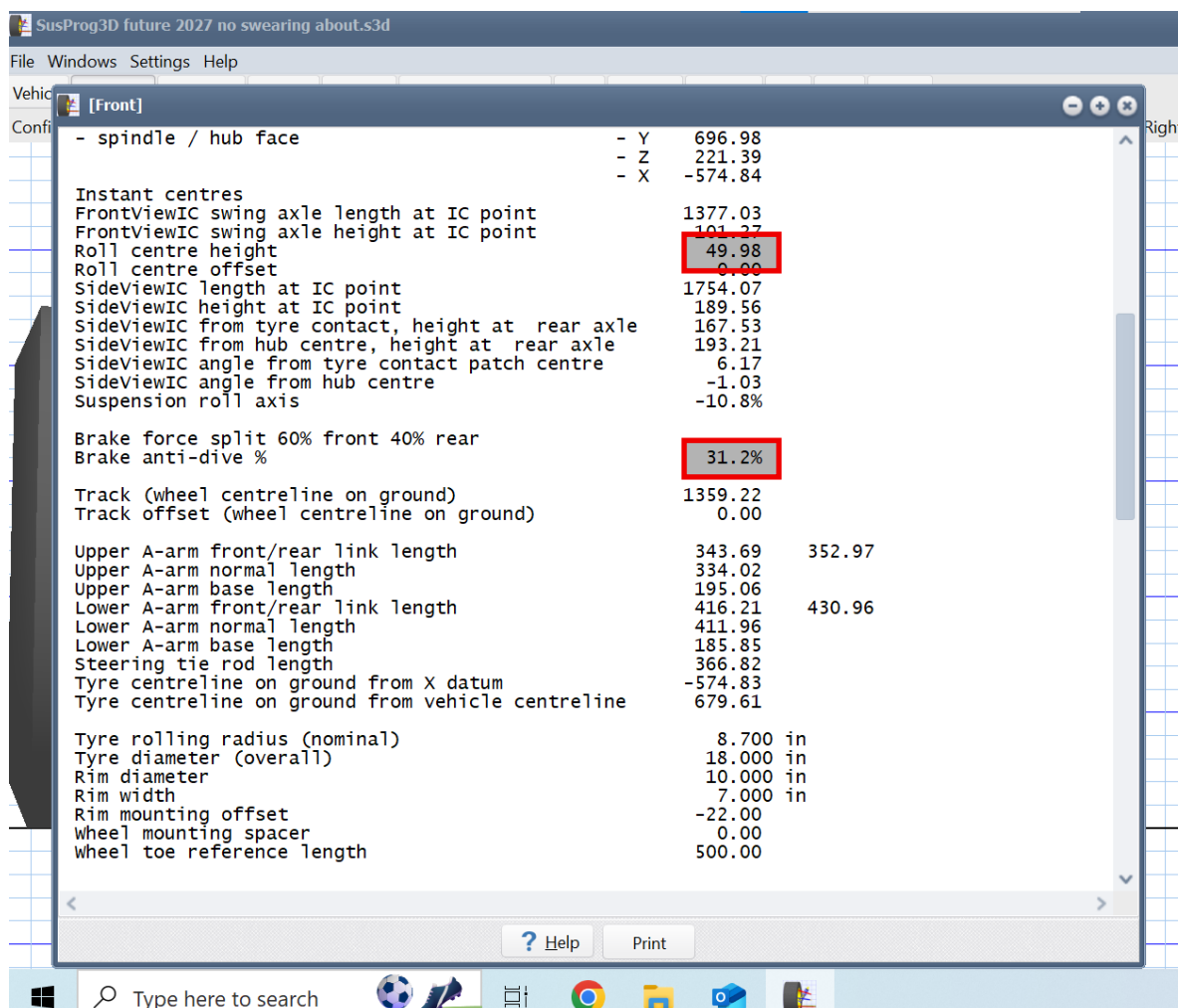


Figure 37 calculated "geometry" results

Initial Design Idea

(fig. 38/39): for the front suspension a bell-crank was applied. Compared to the 2025 DMU FS car, this is a new addition. A bell-crank allows for more adjustability for the coil-over placement. A bell-crank transfers the force from the push rod to the coil-over which would normally be lost in this setup as the coil-over compresses at a completely different angle than the angle the push rod is applying a force.

This allows the coil-over to be mounted on the exterior of the chassis. This still allows for enough space to be left for the driver's legs, pedals and other key components towards the front of the car since it doesn't take up interior space.

Furthermore, the bell-crank means the pushrod arm angle can be adjusted to mitigate bending forces. Having the pushrod more horizontally angled means some of the force transferred by the wheels pushing up is applied directly into bending the rod, this can cause critical failure. Since there is a bell-crank the pushrod can be setup more vertically, this is because the bell-crank can control the motion ratio instead of the pushrod angle.

Motion ratio is how much the wheels move in comparison to how much the coil-over compresses. This is used to achieve a target value for the Hz of the spring which can affect the handling of the car. The length of the arms on the bell-crank that connect to the pushrod and coil-over can be adjusted to increase or decrease the motion ratio.

In (fig. 38): red outline for pushrod, yellow for coil-over, grey for bell crank.

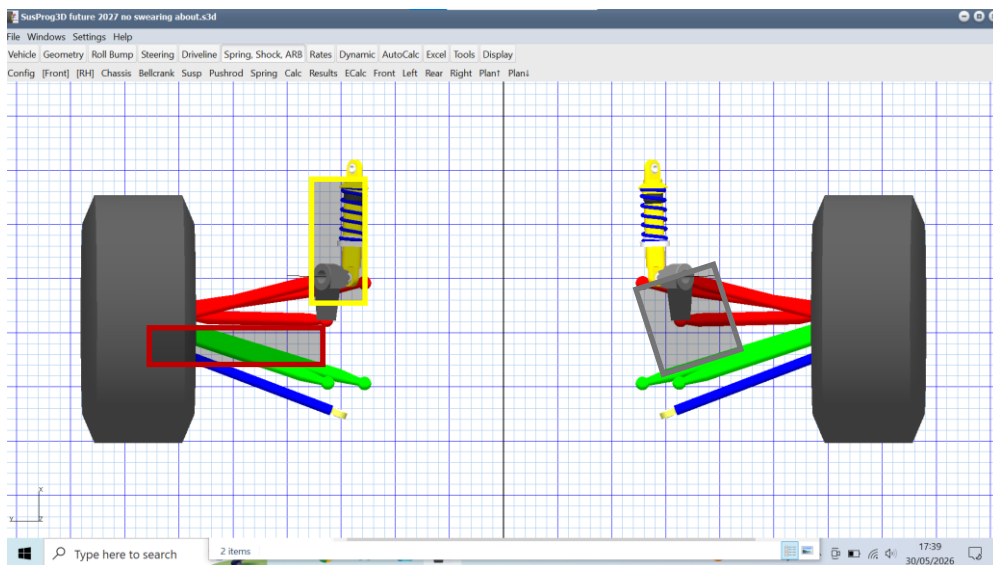


Figure 38 finalised front suspension geometry plan view

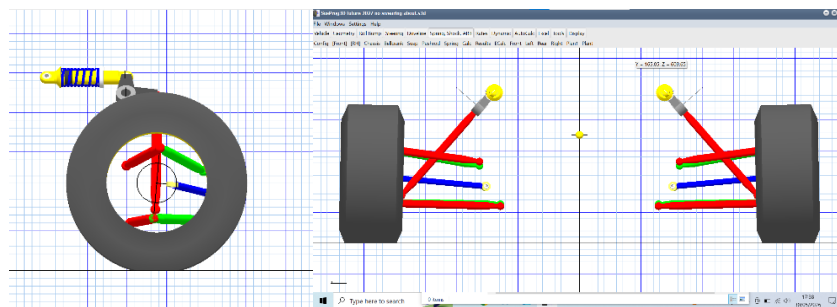


Figure 39 finalised front suspension geometry left side view & front view respectively

(fig. 40/41): The rear suspension geometry was kept similar to the past suspension used for the DMU FS car. A pushrod to bell-crank to coil-over. The aforementioned system was lowered however, to lower the overall centre of gravity. Although this would be a small change, $<1\%$, many small differences add up, meaning the future improvements done by DMU's FS team will add up to make a bigger difference.

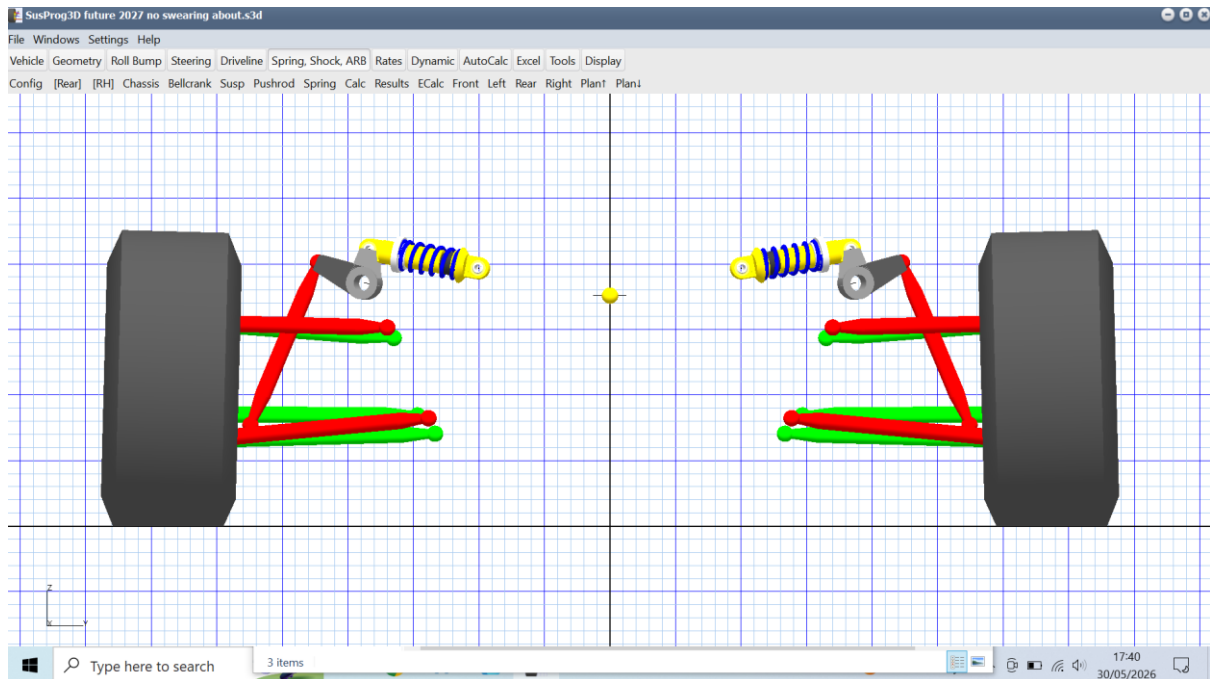


Figure 40 finalised rear suspension geometry front view

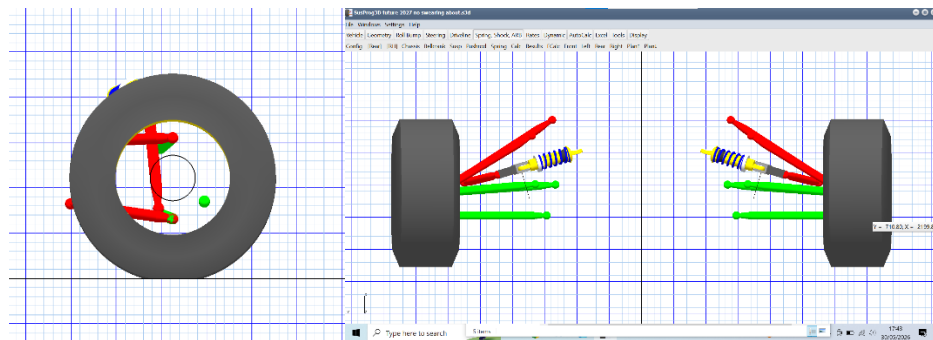


Figure 41 rear suspension geometry left side & plan view respectively

Track Length

(fig. 42): the track length of the rear suspension was much different than the front. This can be visually realised by looking at the distance of the wheels and the screen's borders compared to that of (fig. 43). This can then be confirmed under the alignment tab of the geometry section (highlighted in green). In the alignment tab, (highlighted in red), the value of half the track is shown to be 629.99mm. However, as shown in (fig. 43), the front suspension value is 679.92mm.

The rear suspension having a small track can cause space issues as the drivetrain and engine are placed in the back of the car. Having a significantly smaller track in the rear suspension compared to the front is also problematic as this would increase the car's inclination to oversteer and reduce rear grip.

Although some oversteer could be beneficial at a track like FS Silverstone due to the extremely narrow and sharp corners the car must go through. So, a rear track smaller than the front track by about a few millimetres could be advantageous.

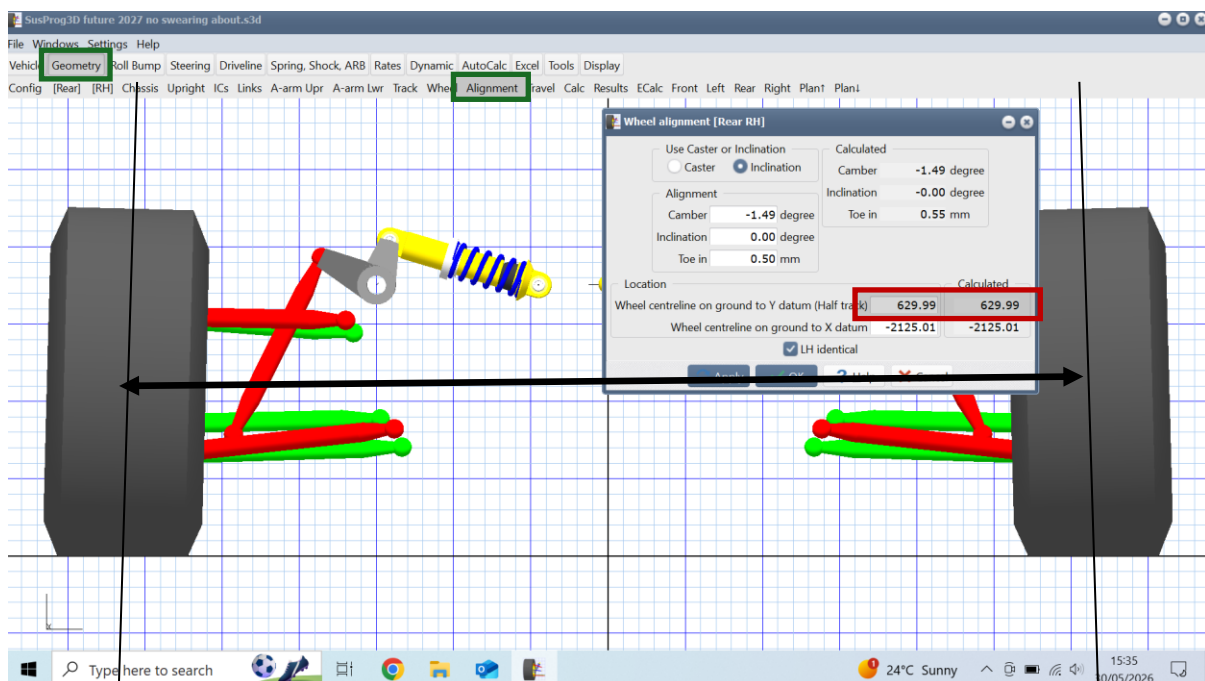


Figure 42 rear suspension geometry before finalisation, front view

(fig. 44): the half-track length has now been changed to 675 to make the full track length 10mm smaller than the front.

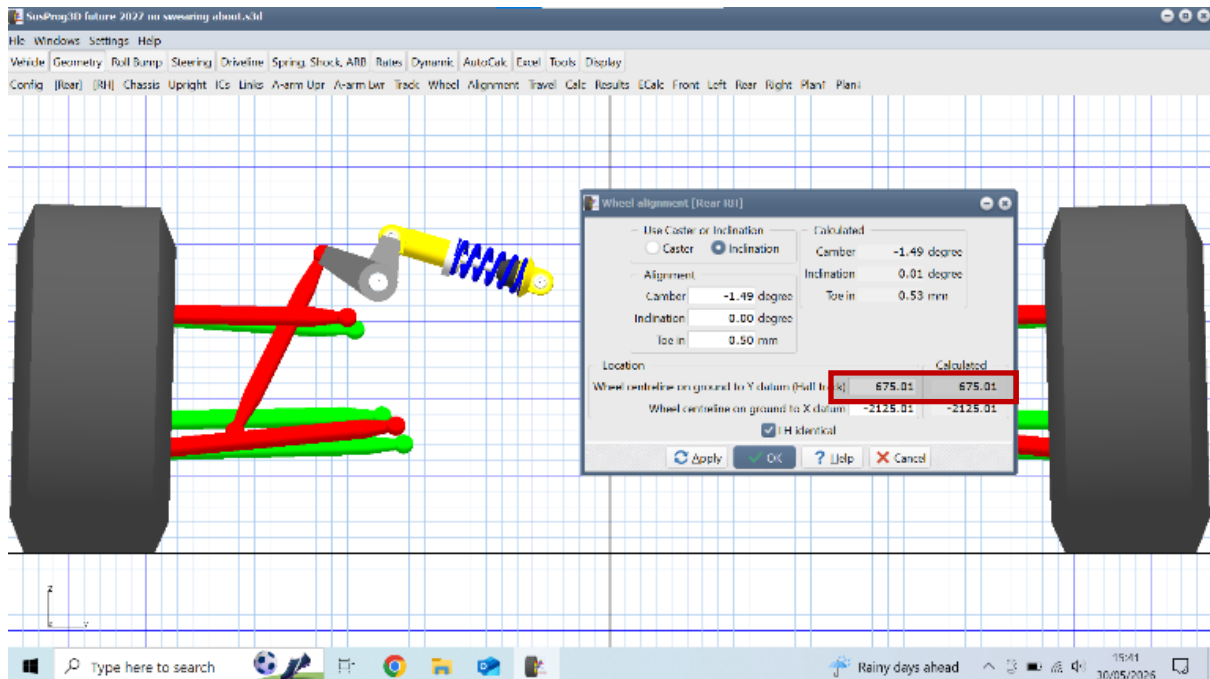


Figure 43 rear geometry after changing half-track length to 675mm

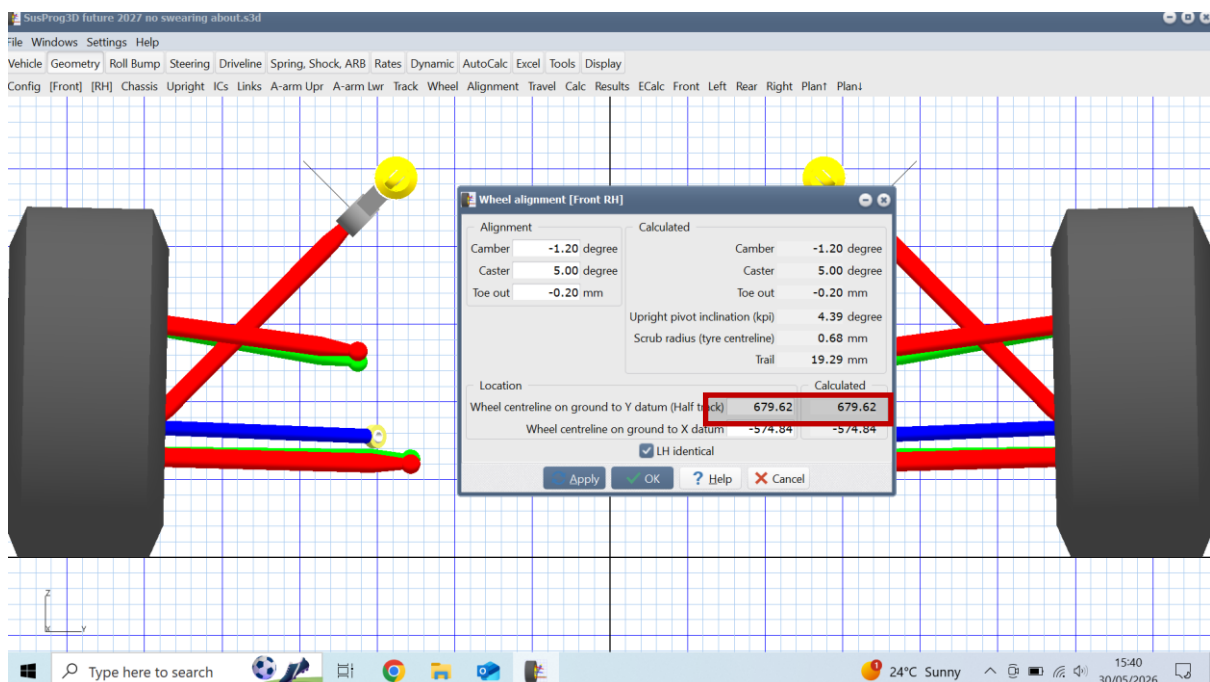


Figure 44 front suspension alignment settings

Front Geometry

See fig. 48 of the appendix

The front suspension uses a double wishbone design, where two A-shaped arms connect each wheel to the chassis and control how the wheel moves under all driving condition. The wheels are set with a slight inward lean of 1.2 degrees (negative camber) at rest, which compensates for the natural outward lean of the car during cornering, keeping the tyre flat against the road where it generates the most grip. A tiny toe-out of 0.2mm – meaning the wheels point very slightly outward – improves the car's responsiveness when turning into a corner. The front geometry is also designed to resist the nose of the car dipping under heavy braking by 31.2%, a consistent value through the full suspension travel range, which ensures stable and predictable braking behaviour regardless of road surface conditions.

Rear Geometry

See fig. 49 of the appendix

The rear suspension also uses a double wishbone design, with an additional toe control link that keeps the rear wheels pointing in a consistent and controlled direction throughout all suspension's movement. The rear wheels are set with 0.28mm of toe-in, meaning they point very slightly inward, which promotes straight-line stability and prevents the rear of the car from sliding outward during cornering. The rear geometry is designed so that 51% of the force that would otherwise cause the rear of the car to squat under acceleration is resisted by the suspension geometry itself, keeping the car level and maintain consistent tyre contact with the road when acceleration out of corners. Additionally, 24.5% of the rear lifting force under braking is resisted geometrically, helping the car stay balanced and stable during deceleration.

Front Roll & Bump

See fig. 50 of the appendix

As the car corners and leans to one side, the outer wheel progressively gains negative camber – from 1.2 degrees at rest to 343 degrees at maximum lean of 4 degrees. This progressive change is intentional and ensures the tyre remains as flat as possible against the road surface through the corner, maximising grip. The point around which the car rolls (the roll centre) sits at 49.9mm above the ground and remains stable throughout cornering, varying only slightly between 45mm and 50mm, which means the handling balance stays consistent and predictable. The front suspension resists brake dive by approximately 31% consistently throughout its full range of travel, providing reliable and repeatable braking behaviour at all suspensions positions.

Rear Roll & Bump

See fig. 51 of the appendix

The rear suspension maintains consistent and progressive camber gain during cornering, increasing from 1.49 degrees at rest to 4.14 degrees at maximum lean, keeping the rear tyres well loaded and in good contact with the road throughout corners. The rear roll centre sits at 61.79mm above the ground, slightly higher than the front roll centre of 49.9mm. This difference is deliberate – having the rear roll centre higher than the front creates a rearward tilt to the car's roll axis which encourages the car to rotate more naturally into corners and reduces the tendency for the front to push wide, known as understeer. The roll centre remains stable during cornering, varying only between 52mm and 62mm across the full 4 degrees of chassis roll, ensuring the handling balance remains predictable throughout a corner.

Front Steering

See fig. 52 of the appendix

The front steering uses a rack and pinion system where turning the steering wheel moves a rack left or right, which in turn pulls or pushes the wheels to change direction. The bump steer – any unintended

change in the direction the wheels point as the suspension move up and down – has been reduced to an almost completely flat 0.22 mm throughout the entire suspension travel range, meaning the car travels in a consistent and predictable direction regardless of road surface variation. The Ackermann geometry, which controls the relationship between how much the inner and outer wheels turn in a corner, is positive throughout the steering range at values between 48% and 70%, meaning the inside wheel always turns more than the outside wheel. This is well suited to the low and medium speed corners typical of Formula Student competition, as it reduces tyre scrubbing and improves manoeuvrability. The steering wheel turns 0.82 times from centre to full lock, give a direct and responsive steering feel appropriate for a lightweight race car.

Rear Steering

See fig. 53 of the appendix

The rear bump steer results are exceptionally well optimised – the change in the direction the rear wheels point throughout the full suspension travel range is 0.00mm relative to the static position. This means the rear wheels maintain a completely consistent direction regardless of whether the suspension is compress or extended, which is critical for a race car as any unintended rear toe change during cornering or over bumps can cause sudden and dangerous oversteer. The static toe-in setting of 0.48mm is maintained perfectly uniformly throughout the entire travel range, confirming that the geometry of the rear toe control link has been positioned correctly to eliminate any bump-induced steering effect at the rear axle

Front Spring, shock, ARB

See fig. 54 of the appendix

The front spring and damper assembly use an Ohlin's coil-over with a spring rate 43.79 N/mm (250 lb/in), connected to the suspension via a pushrod and bell crank rocker system. This arrangement allows the spring and damper to be mounted inboard on the chassis rather than directly at the wheel, which reduces the weight of the unsprung component – the parts that move with the wheel – improving the suspension's ability to follow road surface variations quickly and maintain tyre contact. The front suspensions frequency of 2.31 Hz describes how quickly the car bounces up and down and falls within the target range for a Formula Student race car, providing sufficient stiffness for responsive handling while remaining compliant enough to absorb bumps. Each front corner of the car carries a total weight of 71.42kg (this includes the driver in car), split between 16kg of unsprung mass at the wheel and 55.4kg of sprung mass supported by the spring.

Rear Spring, shock, ARB

See fig. 55 of the appendix

The rear spring and damper assembly use the same Ohlin's coil over specifications as the front, with spring rate of 43.79 N/mm, connected via a pushrod and bell crank system. The rear suspensions frequency of 2.00 Hz is slightly lower than the front's 2.31 Hz, which is a deliberate design choice – having the rear bounce more slowly than the front creates a natural pitching motion that helps the car rotate into corners and improves stability under acceleration. Each rear corner carries a total weight of 74.32kg, slightly more than the front's 71.42kg, resulting in a weight distribution of approximately 49% front and 51% rear, which is close to the ideal balanced distribution for a race car and contributes to neutral and predictable handling behaviour.

Dynamics

See fig. 56 of the appendix

The overall dynamic behaviour of the suspension is characterised by front and rear bounce frequencies of 2.31 Hz and 2.00 Hz respectively, both within the recommended range for a Formula Student race car. These frequencies balance the competing requirements of stiffness for responsive handling and compliance for maintaining tyre contact over uneven surfaces. The roll stiffness – the

resistance to the car leaning in corners - is distributed with 58.8% at the front and 41.2% at the rear. This front-biased roll stiffness distribution means the front of the car resists leaning more strongly than the rear, which promotes a consistent and manageable handling balance. The centre of gravity sits 322mm above the ground, and at 1G of lateral acceleration in cornering the simulation shows the car transfers load appropriately between the inner and outer wheels, confirming the suspensions geometry produces a stable and balanced cornering platform.

Figure 46 excel spreadsheet of co-ordinates according to vehicle origin. part 2

Coil over				
	Y	Z	X	
Chassis	275	488.62	-296.91	
Coil over				
	Y	Z	X	
Chassis	200	393.27	-1997.6	

Figure 47 excel spreadsheet of co-ordinates according to vehicle origin. part 3

Conclusion

The primary design decision of this project was the introduction of a bell crank system at the front suspension geometry, replacing a conventional direct pushrod-to-coil over setup. This proved to be a significant improvement in both performance and practicality. In a direct pushrod arrangement, the coil over must be oriented in a compromise position to accommodate the pushrod angle, limiting where it can be mounted and forcing the pushrod into a more horizontal orientation that introduces greater bending loads into the rod – a potential failure point under dynamic racing conditions. The bell crank eliminates this constraint entirely. By transferring and redirecting the pushrod force through a pivoting rocker, the pushrod can be oriented more vertically to minimise bending stress, while the coil over is mounted in board on the chassis rather than at the wheel. This relocates the entire coil over and spring assembly – approximately 560g – from the unsprung to the sprung mass, representing an estimated 3.5% reduction in unsprung corner mass compared to a direct outboard setup. This directly improves the wheel's ability to follow road surface variations rapidly and maintain tyre contact with the ground, which is critical for consistent grip on the technical, low-speed circuit layout typical of Formula Student competition.

The bell crank also enables independent control of the motion ratio through adjustment of its arm lengths, decoupling spring frequency tuning from pushrod geometry – a significant advantage over a direct setup where the two are inherently linked. This resulted in a front spring frequency of 2.31 Hz, within the recommended Formula Student target range, without requiring the pushrod angle to be compromised to achieve it.

The disadvantages of the bell crank must be acknowledged. It introduces additional components – the rocker itself, its pivot bearing, and additional mounting hardware – which adds complexity and potential failure points compared to a simpler direct system. There is also a small weight penalty associated with these extra components, and manufacturing and assembly tolerances become more critical, as any play in the bell crank pivot will introduce compliance into the system. These are manageable drawbacks for a competition vehicle but must be carefully considered during the manufacturing phase by the 2027 DMU FS team.

Beyond the bell crank decision, the geometry as a whole was designed with careful consideration of the key alignment parameters and their dynamic effects on car behaviour. The front suspension was set with 1.23 degrees of negative camber at rest, a deliberate choice to counteract the positive camber induced by body roll during cornering, ensuring the tyre contact patch remains as flat as possible against the road surface where grip is generated. As the chassis rolls up to 4 degrees, the outer wheel progressively gains negative camber – reaching 3.43 degrees – demonstrating that the geometry actively compensated for cornering loads rather than simply tolerating them. Similarly, a front toe-out of 0.2mm improves corner entry responsiveness, while the rear toe-in of 0.28mm promotes straight-line stability and resistance to oversteer under power. The rear roll centre at 61.99mm sits deliberately higher than the front at 49.9mm, creating a rearward tilt to the car's roll axis that encourages natural rotation into corners and reduces understeer – a handling characteristic well suited to the tight, technical nature of the Formula Student Silverstone circuit. Bump steer was reduced to a peak of 0.02mm at the front and almost eliminated entirely at the rear at a peak of 0.01, ensuring that road surface variation does not introduce unintended steering inputs into either axle – a criterion that Formula Student design judges explicitly evaluate.

Taken together, the suspension geometry produced by this project represents a well-considered and data-driven baseline for the DMU 2027 Formula Student car. The susprog3d outputs, coordinates, and component specifications compiled here provide the future team with a ready foundation from which 3D modelling, FEA analysis, and manufacturing can proceed.

Design Specifications

Design Specification			
Parameter	Requirement	Achieved	Evidence
Susprog3d Geometry	Co-ordinates	yes	Excel spreadsheet of finalised geometry coordinates <i>Page 41-42, fig. 45-47</i>
Dimensions - Track	No more than 75% difference between front and rear	yes	Changing track to more optimal range <i>Page 36-37, fig. 42-44</i>
Dimensions - Wheelbase	Minimum 1525mm	yes	<i>Appendix, wheelbase length susprog3d section image</i> <i>Page 60, fig. 58</i>
Clearance	30mm above ground	yes	<i>Appendix, ride height susprog3d section image</i> <i>Page 59, fig. 57</i>
Mass	Improved unsprang mass	yes	Front Bell crank <i>Page 34, fig. 38</i>

References

- [1] “United Nations 17 sustainable development goals”. *Un.org*, 2015 <https://sdgs.un.org/goals> (March. 14, 2026).
- [2] “On the Influence of Suspension Geometry on Steering Feedback”. *Uni of Brescia*, 2020 [On the Influence of Suspension Geometry on Steering Feedback | MDPI](#) (March. 15, 2026).
- [3] “Design & Analysis of Formula Student Race Suspension System”. *Procedia Engineering*, 2016 <https://www.sciencedirect.com/science/article/pii/S1877705816302983> (March. 16, 2026).
- [4] “Race Car Design”. *Derek Seward*, 2014 <https://books.google.co.uk/books?hl=en&lr=&id=s3atEAAQBAJ&oi=fnd&pg=PP1&dq=racecar+design&ots=XIO7UIFsDs&sig=FQtzxRBklyO00IIGiTMhc-mNRak#v=onepage&q=racecar%20design&f=false> (March. 20, 2026).
- [5] “Design Analysis & Optimisation of Race Car Suspension”. *University of Brasov*, 2021 <https://www.mdpi.com/2076-3417/11/9/4167> (March. 20, 2026).
- [6] “Formula Student Rules & Regulations”. IMechE, 2026 <https://www.imeche.org/docs/default-source/1-oscar/formula-student/2026/rules/fsuk-2026-rules---v1-09e21118e54216d0c8310ff0100d05193.pdf?sfvrsn=2> (March. 22, 2026).

Appendix

SusProg3D future 2027 no swearing about.s3d Front Geometry

Double A-arm, steering link

Vehicle horizontal datum (Y): Vehicle centreline
Vehicle vertical datum (Z): Ground
Vehicle longitudinal datum (X): Bulkhead

Chassis pivot points (from vehicle Y, Z, X datum)

- upper A-arm chassis pivot (front/rear)	- Y	325.00	325.00
	- Z	264.99	251.33
	- X	-498.33	-692.91
- lower A-arm chassis pivot (front/rear)	- Y	257.00	257.00
	- Z	120.06	126.33
	- X	-508.02	-693.76
- tie rod	- Y	310.00	
	- Z	183.56	
	- X	-749.01	

Upright pivot points (from vehicle Y, Z, X datum)

- upper A-arm upright pivot	- Y	655.10	
	- Z	310.19	
	- X	-582.66	
- lower A-arm upright pivot	- Y	668.86	
	- Z	131.13	
	- X	-566.99	
- tie rod (steering arm)	- Y	657.07	
	- Z	215.33	
	- X	-634.59	
- spindle / wheel cl	- Y	674.98	
	- Z	220.93	
	- X	-574.83	
- spindle / hub face	- Y	696.98	
	- Z	221.39	
	- X	-574.84	

Instant centres

FrontViewIC swing axle length at IC point	1377.03
FrontViewIC swing axle height at IC point	101.27
Roll centre height	49.98
Roll centre offset	0.00
SideViewIC length at IC point	1754.07
SideViewIC height at IC point	189.56
SideViewIC from tyre contact, height at rear axle	167.53
SideViewIC from hub centre, height at rear axle	193.21
SideViewIC angle from tyre contact patch centre	6.17
SideViewIC angle from hub centre	-1.03
Suspension roll axis	-10.8%

Brake force split 60% front 40% rear

Brake anti-dive % 31.2%

Track (wheel centreline on ground) 1359.22

Track offset (wheel centreline on ground) 0.00

Upper A-arm front/rear link length	343.69	352.97
Upper A-arm normal length	334.02	
Upper A-arm base length	195.06	
Lower A-arm front/rear link length	416.21	430.96
Lower A-arm normal length	411.96	
Lower A-arm base length	185.85	
Steering tie rod length	366.82	
Tyre centreline on ground from X datum	-574.83	
Tyre centreline on ground from vehicle centreline	679.61	

Tyre rolling radius (nominal)	8.700 in
Tyre diameter (overall)	18.000 in
Rim diameter	10.000 in
Rim width	7.000 in
Rim mounting offset	-22.00
Wheel mounting spacer	0.00
Wheel toe reference length	500.00

Wheel alignment in straight ahead position

Camber angle	-1.20
Upper kingpin point (actual)	- Y 655.10
	- Z 310.19
	- X -582.66
Lower kingpin point (actual)	- Y 668.86
	- Z 131.13
	- X -566.99

Upright pivot inclination (kpi)

Upright pivot scrub radius (tyre cl on ground)	0.68	
Caster angle and trail	5.00	19.32
Steering offset	19.33	
Spindle offset from kingpin axis (side/front view)	0.00	12.98
Toe out (mm)	-0.20	

Ride height ref point (from vehicle Y, Z, X datum)	Front LH	Front RH	Rear LH	Rear RH
- Y	-208.00	208.00	-225.00	225.00
Ride height (ref point to ground)	- Z 40.00	40.00	50.00	50.00
	- X -499.99	-499.99	-1959.95	-1959.95

Datum reference dimensions

Chassis horizontal datum (Y): Chassis centreline
Chassis vertical datum (Z): Ground
Chassis longitudinal datum (X): Bulkhead

Ride height ref point (from chassis Y, Z, X datum) Front LH Rear LH

Calculated using SusProg3D V5.160R (1741.17) by DMURacing_Laptop 30/05/2026 18:29

	- Y	-208.00	-225.00
	- Z	0.00	0.00
	- X	-500.00	-1960.00
Chassis pivot points (from chassis Y, Z, X datum)			
- upper A-arm (front/rear)	- Y	325.00	325.00
	- Z	225.00	210.00
	- X	-499.88	-694.36
- lower A-arm (front/rear)	- Y	257.00	257.00
	- Z	80.00	85.00
	- X	-508.58	-694.36
- tie rod	- Y	310.00	
	- Z	141.85	
	- X	-750.00	
Upright pivot points (from upright Y, Z, X datum)			
Camber shim adjusts both top pivot and toe control pivot			
- camber shim stack (included in Y dimensions)		10.00	
- camber shim thickness (one shim)		2.00	
- camber shim count		5	
- upper A-arm	- Y	40.00	
	- Z	90.00	
	- X	0.00	
- lower A-arm	- Y	30.00	
	- Z	-90.00	
	- X	0.00	
- tie rod (steering arm)	- Y	40.00	
	- Z	0.00	
	- X	-60.00	
Upright pivot points (from chassis Y, Z, X datum)			
- upper A-arm	- Y	655.10	
	- Z	269.62	
	- X	-584.52	
- lower A-arm	- Y	668.86	
	- Z	90.66	
	- X	-567.62	
- tie rod (steering arm)	- Y	657.07	
	- Z	174.40	
	- X	-635.80	

Calculated 30 May 2026 using SusProg3D V5.160R (1741.17)

Figure 48 susprog3d front suspension "geometry" section results

2027 NO SWEARING ABOUT SSD REAR GEOMETRY

Double A-arm, toe link

Vehicle horizontal datum (Y): Vehicle centreline
Vehicle vertical datum (Z): Ground
Vehicle longitudinal datum (X): Bulkhead

Chassis pivot points (from chassis Y, Z, X datum)	LH		RH	
- upper A-arm chassis pivot (front/rear)	- Y	340.00	330.00	-340.00
	- Z	302.93	286.95	302.93
	- X	-1948.22	-2098.33	-1948.22
- lower A-arm chassis pivot (front/rear)	- Y	277.00	267.00	-277.00
	- Z	164.59	140.96	164.59
	- X	-1899.17	-2099.33	-1899.17
- toe control link	- Y	294.00		-294.00
	- Z	168.86		168.86
	- X	-2195.06		-2195.06
Upright pivot points (from vehicle Y, Z, X datum)				
- upper A-arm upright pivot	- Y	634.04		-634.04
	- Z	310.02		310.02
	- X	-2125.13		-2125.13
- lower A-arm upright pivot	- Y	626.69		-626.69
	- Z	131.35		131.35
	- X	-2125.14		-2125.14
- toe control link upright pivot	- Y	652.75		-652.75
	- Z	170.45		170.45
	- X	-2195.11		-2195.11
- spindle / wheel cl	- Y	669.37		-669.37
	- Z	220.91		220.91
	- X	-2125.10		-2125.10
- spindle / hub face	- Y	691.36		-691.36
	- Z	221.48		221.48
	- X	-2125.07		-2125.07
Instant centres				
FrontViewIC swing axle length at IC point		1778.71		1778.70
FrontViewIC swing axle height at IC point		162.78		162.78
roll centre height			61.79	
roll centre offset			0.00	
SideViewIC length at IC point		10500.30		10500.30
SideViewIC height at IC point		1335.34		1335.34
SideViewIC from tyre contact, height at front axle		197.15		197.15
SideViewIC from hub centre, height at front axle		385.50		385.50
SideViewIC angle from tyre contact patch centre		7.25		7.25
SideViewIC angle from hub centre		6.06		6.06
suspension roll axis			12.6%	
brake force split 60% front 40% rear				
brake anti-lift %		24.5%		24.5%
acceleration anti-squat %		51.0%		51.0%
Track (wheel centreline on ground)		1350.24		
Track offset (wheel centreline on ground)		0.00		
Upper A-arm front/rear link length		343.23	306.09	343.23
Upper A-arm normal length		306.06		306.06
Upper A-arm base length		151.29		151.29
Lower A-arm front/rear link length		417.67	360.74	417.67
Lower A-arm normal length		360.63		360.63
Lower A-arm base length		201.80		201.80
Toe control link length		358.75		358.75
Tyre centreline on ground from X datum		-2125.09		-2125.09
Tyre centreline on ground from vehicle centreline		675.12		-675.12
Tyre rolling radius (nominal)		8.700 in		8.700 in
Tyre diameter (overall)		18.000 in		18.000 in
Rim diameter		10.000 in		10.000 in
Rim width		7.000 in		7.000 in
Rim mounting offset		-22.00		-22.00
Wheel mounting spacer		0.00		0.00
Wheel toe reference length		500.00		500.00
Wheel alignment in straight ahead position				
Camber angle		-1.49		-1.49
Inclination angle		0.00		0.00
LH toe in RH toe in (mm)		0.48		0.48

Ride height (ref point to ground)	- Y	-208.00	208.00	-225.00	225.00
	- Z	40.00	40.00	50.00	50.00
	- X	-499.99	-499.99	-1959.95	-1959.95
Datum reference dimensions					
Chassis horizontal datum (Y): Chassis centreline					
Chassis vertical datum (Z): Ground					
Chassis longitudinal datum (X): Bulkhead					
Ride height ref point (from chassis Y, Z, X datum)			Front LH		Rear LH
	- Y	-208.00		-225.00	
	- Z	0.00		0.00	
	- X	-500.00		-1960.00	
Chassis pivot points (from chassis Y, Z, X datum)					
- upper A-arm (front/rear)	- Y	340.00	330.00		
	- Z	253.00	236.00		
	- X	-1950.00	-2100.00		
- lower A-arm (front/rear)	- Y	277.00	267.00		
Calculated using SusProg3D V5.160R (1741.17) by DMURacing Laptop 30/05/2026 17:58					
	- Z	115.00	90.00		
	- X	-1900.00	-2100.00		
- toe control link	- Y	294.00			
	- Z	117.25			
	- X	-2195.91			
Upright pivot points (from upright Y, Z, X datum)					
Camber shim adjusts top pivot only					
- camber shim stack (included in Y dimensions)		5.00			
- camber shim thickness (one shim)		1.00			
- camber shim count		5			
- upper A-arm	- Y	55.00			
	- Z	90.00			
	- X	0.00			
- lower A-arm	- Y	67.00			
	- Z	-88.41			
	- X	0.00			
- toe control link	- Y	40.00			
	- Z	-50.00			
	- X	-70.00			
Upright pivot points (from chassis Y, Z, X datum)					
- upper A-arm	- Y	634.04			
	- Z	258.88			
	- X	-2126.95			
- lower A-arm	- Y	626.69			
	- Z	80.22			
	- X	-2125.74			
- toe control link	- Y	652.75			
	- Z	118.84			
	- X	-2195.98			
Calculated 30 May 2026 using SusProg3D V5.160R (1741.17)					

Figure 49 susprog3d rear suspension "geometry" section results

susprog3D future 2027 no swearing about.ssd front roll and bump
 Chassis roll values calculated every 0.25 degrees
 SemiDynamic roll centre, horizontal roll axis. Roll starts at Static.
 Toe variation has NOT been calculated.

LH wheel	camber angle	caster angle	caster trail	kpi angle	scrub radius	steering offset	wheel scrub	axle tramp	toe mm	roll centz offset	centz height
4.00 roll LH	0.62	6.54	25.34	2.58	0.64	25.35	-0.51	-0.82	-0.20	-218.77	45.4
3.50 roll LH	0.42	6.35	24.59	2.78	0.65	24.60	-0.41	-0.72	-0.20	-192.43	46.4
3.00 roll LH	0.21	6.16	23.84	2.99	0.65	23.85	-0.31	-0.62	-0.20	-165.68	47.3
2.50 roll LH	-0.01	5.96	23.09	3.21	0.66	23.10	-0.22	-0.51	-0.20	-138.60	48.1
2.00 roll LH	-0.24	5.77	22.33	3.43	0.66	22.34	-0.15	-0.41	-0.20	-111.23	48.6
1.50 roll LH	-0.47	5.58	21.58	3.66	0.67	21.59	-0.09	-0.31	-0.20	-83.62	49.3
1.00 roll LH	-0.70	5.39	20.83	3.90	0.67	20.84	-0.04	-0.21	-0.20	-55.85	49.6
0.50 roll LH	-0.95	5.20	20.07	4.14	0.67	20.08	-0.01	-0.10	-0.20	-27.95	49.9
0.00 roll LH	-1.20	5.00	19.32	4.39	0.68	19.33	0.00	0.00	-0.20	0.00	49.9
0.50 roll RH	-1.46	4.81	18.57	4.65	0.68	18.58	-0.01	0.10	-0.20	27.95	49.9
1.00 roll RH	-1.72	4.62	17.81	4.91	0.68	17.83	-0.03	0.21	-0.20	55.85	49.6
1.50 roll RH	-1.99	4.43	17.06	5.18	0.69	17.08	-0.08	0.31	-0.20	83.62	49.3
2.00 roll RH	-2.27	4.23	16.31	5.45	0.69	16.32	-0.15	0.42	-0.20	111.23	48.6
2.50 roll RH	-2.55	4.04	15.56	5.74	0.70	15.57	-0.24	0.52	-0.20	138.60	48.1
3.00 roll RH	-2.83	3.85	14.81	6.02	0.70	14.82	-0.36	0.62	-0.20	165.68	47.3
3.50 roll RH	-3.13	3.66	14.06	6.31	0.70	14.08	-0.51	0.73	-0.20	192.43	46.4
4.00 roll RH	-3.43	3.47	13.31	6.61	0.70	13.33	-0.68	0.83	-0.20	218.77	45.4
RH wheel	camber angle	caster angle	caster trail	kpi angle	scrub radius	steering offset	wheel scrub	axle tramp	toe mm	roll centz offset	centz height
4.00 roll LH	-3.43	3.47	13.31	6.61	0.70	13.33	-0.68	0.83	-0.20	-218.77	45.4
3.50 roll LH	-3.13	3.66	14.06	6.31	0.70	14.08	-0.51	0.73	-0.20	-192.43	46.4
3.00 roll LH	-2.83	3.85	14.81	6.02	0.70	14.82	-0.36	0.62	-0.20	-165.68	47.3
2.50 roll LH	-2.55	4.04	15.56	5.74	0.70	15.57	-0.24	0.52	-0.20	-138.60	48.1
2.00 roll LH	-2.27	4.23	16.31	5.45	0.69	16.32	-0.15	0.42	-0.20	-111.23	48.6
1.50 roll LH	-1.99	4.43	17.06	5.18	0.69	17.08	-0.08	0.31	-0.20	-83.62	49.3
1.00 roll LH	-1.72	4.62	17.81	4.91	0.68	17.83	-0.03	0.21	-0.20	-55.85	49.6
0.50 roll LH	-1.46	4.81	18.57	4.65	0.68	18.58	-0.01	0.10	-0.20	-27.95	49.9
0.00 roll LH	-1.20	5.00	19.32	4.39	0.68	19.33	0.00	0.00	-0.20	0.00	49.9
0.50 roll RH	-0.95	5.20	20.07	4.14	0.67	20.08	-0.01	-0.10	-0.20	27.95	49.9
1.00 roll RH	-0.70	5.39	20.83	3.90	0.67	20.84	-0.04	-0.21	-0.20	55.85	49.6
1.50 roll RH	-0.47	5.58	21.58	3.66	0.67	21.59	-0.09	-0.31	-0.20	83.62	49.3
2.00 roll RH	-0.24	5.77	22.33	3.43	0.66	22.34	-0.15	-0.41	-0.20	111.23	48.6
2.50 roll RH	-0.01	5.96	23.09	3.21	0.66	23.10	-0.22	-0.51	-0.20	138.60	48.1
3.00 roll RH	0.21	6.16	23.84	2.99	0.65	23.85	-0.31	-0.62	-0.20	165.68	47.3
3.50 roll RH	0.42	6.35	24.59	2.78	0.65	24.60	-0.41	-0.72	-0.20	192.43	46.4
4.00 roll RH	0.62	6.54	25.34	2.58	0.64	25.35	-0.51	-0.82	-0.20	218.77	45.4
LH and RH wheel	camber angle	caster angle	caster trail	kpi angle	scrub radius	steering offset	wheel scrub	axle tramp	toe mm	rc offset	roll ce chassi
26.00 bump	-2.34	5.86	22.62	5.54	0.66	22.63	1.22	-0.45	-0.20	0.00	41.2
25.00 bump	-2.29	5.83	22.49	5.49	0.66	22.50	1.20	-0.44	-0.20	0.00	41.5
20.00 bump	-2.07	5.67	21.86	5.26	0.67	21.87	1.06	-0.35	-0.20	0.00	43.1
15.00 bump	-1.84	5.50	21.23	5.04	0.67	21.24	0.87	-0.26	-0.20	0.00	44.6
10.00 bump	-1.62	5.33	20.59	4.82	0.67	20.60	0.63	-0.18	-0.20	0.00	46.5
5.00 bump	-1.41	5.17	19.96	4.60	0.67	19.97	0.34	-0.09	-0.20	0.00	48.2
Static	-1.20	5.00	19.32	4.39	0.68	19.33	0.00	0.00	-0.20	0.00	49.9
5.00 droop	-1.00	4.84	18.68	4.19	0.68	18.70	-0.39	0.09	-0.20	0.00	51.7
10.00 droop	-0.79	4.67	18.05	3.98	0.68	18.06	-0.83	0.18	-0.20	0.00	53.4
15.00 droop	-0.60	4.51	17.41	3.79	0.69	17.43	-1.32	0.26	-0.20	0.00	55.2
20.00 droop	-0.40	4.34	16.78	3.59	0.69	16.79	-1.87	0.35	-0.20	0.00	57.0
25.00 droop	-0.22	4.18	16.14	3.40	0.69	16.16	-2.46	0.44	-0.20	0.00	58.6
26.00 droop	-0.18	4.15	16.01	3.37	0.69	16.03	-2.58	0.46	-0.20	0.00	59.2

Equivalent suspension travel due to chassis roll

	LH	RH
4.00 roll LH	47.29	-47.53
3.50 roll LH	41.40	-41.58
3.00 roll LH	35.50	-35.64
2.50 roll LH	29.60	-29.69
2.00 roll LH	23.69	-23.75
1.50 roll LH	17.77	-17.81
1.00 roll LH	11.85	-11.87
0.50 roll LH	5.93	-5.93
0.00 roll	0.00	0.00
0.50 roll RH	-5.93	5.93
1.00 roll RH	-11.87	11.85
1.50 roll RH	-17.81	17.77
2.00 roll RH	-23.75	23.69
2.50 roll RH	-29.69	29.60
3.00 roll RH	-35.64	35.50
3.50 roll RH	-41.58	41.40
4.00 roll RH	-47.53	47.29

Side view swing axle and instant centre

	IC length	IC height	axle height	ground angle	hub angle
26.00 bump	1768.25	189.43	166.08	6.11	-1.02
25.00 bump	1767.57	189.43	166.14	6.12	-1.02
20.00 bump	1764.37	189.41	166.43	6.13	-1.02
15.00 bump	1761.45	189.42	166.71	6.14	-1.03
10.00 bump	1758.73	189.45	166.99	6.15	-1.03
5.00 bump	1756.28	189.49	167.27	6.16	-1.03
Static	1754.07	189.56	167.53	6.17	-1.03
5.00 droop	1752.07	189.65	167.80	6.18	-1.02
10.00 droop	1750.29	189.75	168.06	6.19	-1.02
15.00 droop	1748.70	189.87	168.32	6.20	-1.02
20.00 droop	1747.32	190.01	168.58	6.21	-1.02
25.00 droop	1746.12	190.17	168.83	6.22	-1.01
26.00 droop	1745.90	190.20	168.89	6.22	-1.01

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	LH brake a-dive%	accel a-lift%
26.00 bump	30.9%	0.0%
25.00 bump	30.9%	0.0%
20.00 bump	31.0%	0.0%
15.00 bump	31.0%	0.0%
10.00 bump	31.1%	0.0%
5.00 bump	31.1%	0.0%
Static	31.2%	0.0%
5.00 droop	31.2%	0.0%
10.00 droop	31.3%	0.0%
15.00 droop	31.3%	0.0%
20.00 droop	31.4%	0.0%
25.00 droop	31.4%	0.0%
26.00 droop	31.4%	0.0%

Calculated 31 May 2026 using SusProg3D V5.160R (1741.17)

Figure 50 susprog3d front suspension "roll bump" section results

SusProg3D future 2027 no swearing about.s3d Rear Roll and bump
 Chassis roll values calculated every 0.25 degrees
 SemiDynamic roll centre, horizontal roll axis. Roll starts at Static.
 Toe variation has NOT been calculated.

LH wheel	camber angle	inclin angle	wheel scrub	axle tramp	toe mm	roll centre offset	roll centre height	fvsax	
4.00 roll LH	0.80	-0.29	-0.68	-4.85	0.48	-276.87	52.57	1410.30	
3.50 roll LH	0.54	-0.25	-0.54	-4.27	0.48	-244.02	54.66	1451.23	
3.00 roll LH	0.26	-0.21	-0.42	-3.68	0.48	-210.48	56.51	1493.45	
2.50 roll LH	-0.01	-0.17	-0.30	-3.08	0.48	-176.34	58.10	1537.03	
2.00 roll LH	-0.30	-0.14	-0.20	-2.48	0.48	-141.69	59.41	1582.07	
1.50 roll LH	-0.59	-0.10	-0.12	-1.87	0.48	-106.63	60.44	1628.64	
1.00 roll LH	-0.88	-0.07	-0.05	-1.25	0.48	-71.24	61.19	1676.86	
0.50 roll LH	-1.18	-0.03	-0.01	-0.63	0.48	-35.68	61.64	1726.83	
0.00 roll	-1.49	0.00	0.00	0.00	0.48	0.00	61.79	1778.68	
0.50 roll RH	-1.80	0.03	-0.01	0.64	0.48	35.68	61.64	1832.55	
1.00 roll RH	-2.12	0.06	-0.06	1.28	0.48	71.24	61.19	1888.72	
1.50 roll RH	-2.45	0.09	-0.14	1.92	0.48	106.63	60.44	1947.03	
2.00 roll RH	-2.78	0.12	-0.25	2.58	0.48	141.69	59.41	2007.97	
2.50 roll RH	-3.11	0.15	-0.41	3.24	0.48	176.34	58.10	2071.68	
3.00 roll RH	-3.45	0.18	-0.61	3.90	0.48	210.48	56.51	2138.41	
3.50 roll RH	-3.79	0.20	-0.85	4.57	0.48	244.02	54.66	2208.43	
4.00 roll RH	-4.14	0.23	-1.14	5.24	0.48	276.87	52.57	2282.08	
RH wheel	camber angle	inclin angle	wheel scrub	axle tramp	toe mm	roll centre offset	roll centre height	fvsax	
4.00 roll LH	-4.14	0.23	-1.14	5.24	0.48	-276.87	52.57	2282.08	
3.50 roll LH	-3.79	0.20	-0.85	4.57	0.48	-244.02	54.66	2208.43	
3.00 roll LH	-3.45	0.18	-0.61	3.90	0.48	-210.48	56.51	2138.41	
2.50 roll LH	-3.11	0.15	-0.41	3.24	0.48	-176.34	58.10	2071.68	
2.00 roll LH	-2.78	0.12	-0.25	2.58	0.48	-141.69	59.41	2007.97	
1.50 roll LH	-2.45	0.09	-0.14	1.92	0.48	-106.63	60.44	1947.03	
1.00 roll LH	-2.12	0.06	-0.06	1.28	0.48	-71.24	61.19	1888.72	
0.50 roll LH	-1.80	0.03	-0.01	0.64	0.48	-35.68	61.64	1832.55	
0.00 roll	-1.49	0.00	0.00	0.00	0.48	0.00	61.79	1778.68	
0.50 roll RH	-1.18	-0.03	-0.01	-0.63	0.48	35.68	61.64	1726.83	
1.00 roll RH	-0.88	-0.07	-0.05	-1.25	0.48	71.24	61.19	1676.86	
1.50 roll RH	-0.59	-0.10	-0.12	-1.87	0.48	106.63	60.44	1628.64	
2.00 roll RH	-0.30	-0.14	-0.20	-2.48	0.48	141.69	59.41	1582.07	
2.50 roll RH	-0.01	-0.17	-0.30	-3.08	0.48	176.34	58.10	1537.03	
3.00 roll RH	0.26	-0.21	-0.42	-3.68	0.48	210.48	56.51	1493.45	
3.50 roll RH	0.54	-0.25	-0.54	-4.27	0.48	244.02	54.66	1451.23	
4.00 roll RH	0.80	-0.29	-0.68	-4.85	0.48	276.87	52.57	1410.30	
LH and RH wheel	camber angle	inclin angle	wheel scrub	axle tramp	toe mm	rc offset	roll centre chassis	roll centre height ground	fvsax
31.00 bump	-2.57	-0.19	1.74	-3.22	0.48	0.00	45.32	14.32	1528.85
30.00 bump	-2.53	-0.18	1.72	-3.12	0.48	0.00	45.83	15.83	1536.33
25.00 bump	-2.35	-0.15	1.57	-2.61	0.48	0.00	48.40	23.40	1574.27
20.00 bump	-2.17	-0.12	1.37	-2.10	0.48	0.00	51.00	31.00	1613.12
15.00 bump	-1.99	-0.09	1.11	-1.58	0.48	0.00	53.64	38.64	1652.99
10.00 bump	-1.82	-0.06	0.80	-1.06	0.48	0.00	56.32	46.32	1693.73
5.00 bump	-1.65	-0.03	0.43	-0.53	0.48	0.00	59.03	54.03	1735.63
Static	-1.49	0.00	0.00	0.00	0.48	0.00	61.79	61.79	1778.68
5.00 droop	-1.33	0.03	-0.49	0.54	0.48	0.00	64.58	69.58	1822.94
10.00 droop	-1.18	0.05	-1.03	1.08	0.48	0.00	67.41	77.41	1868.51
15.00 droop	-1.03	0.08	-1.63	1.63	0.48	0.00	70.28	85.28	1915.46
20.00 droop	-0.88	0.11	-2.30	2.18	0.48	0.00	73.19	93.19	1963.90
25.00 droop	-0.73	0.13	-3.02	2.74	0.48	0.00	76.14	101.14	2013.96

Equivalent suspension travel due to chassis roll

	LH	RH
4.00 roll LH	46.95	-47.24
3.50 roll LH	41.10	-41.33
3.00 roll LH	35.25	-35.42
2.50 roll LH	29.39	-29.51
2.00 roll LH	23.52	-23.60
1.50 roll LH	17.65	-17.69
1.00 roll LH	11.77	-11.79
0.50 roll LH	5.89	-5.89
0.00 roll	0.00	0.00
0.50 roll RH	-5.89	5.89
1.00 roll RH	-11.79	11.77
1.50 roll RH	-17.69	17.65
2.00 roll RH	-23.60	23.52
2.50 roll RH	-29.51	29.39
3.00 roll RH	-35.42	35.25
3.50 roll RH	-41.33	41.10
4.00 roll RH	-47.24	46.95

Side view swing axle and instant centre

	IC	IC	axle	ground	hub
	length	height	height	angle	angle
31.00 bump	9275.40	1150.54	192.30	7.07	5.72
30.00 bump	9310.86	1155.89	192.46	7.08	5.73
25.00 bump	9491.39	1183.19	193.25	7.11	5.79
20.00 bump	9677.98	1211.43	194.05	7.13	5.84
15.00 bump	9871.62	1240.74	194.85	7.16	5.90
10.00 bump	10070.60	1270.96	195.65	7.19	5.95
5.00 bump	10277.81	1302.42	196.45	7.22	6.01
Static	10492.92	1335.11	197.25	7.25	6.06
5.00 droop	10716.47	1369.11	198.06	7.28	6.12
10.00 droop	10949.13	1404.54	198.87	7.31	6.17
15.00 droop	11191.62	1441.49	199.68	7.34	6.22
20.00 droop	11444.72	1480.09	200.49	7.37	6.28
25.00 droop	11709.32	1520.48	201.30	7.40	6.33

Calculated using SusProg3D V5.160R (1741.17) by DMURacing_Laptop 30/05/2026 17:51

	brake	accel
	a-lift%	a-squat%
31.00 bump	23.9%	48.2%
30.00 bump	23.9%	48.3%
25.00 bump	24.0%	48.8%
20.00 bump	24.1%	49.2%
15.00 bump	24.2%	49.7%
10.00 bump	24.3%	50.1%
5.00 bump	24.4%	50.6%
Static	24.5%	51.1%
5.00 droop	24.6%	51.5%
10.00 droop	24.7%	52.0%
15.00 droop	24.8%	52.5%
20.00 droop	24.9%	52.9%
25.00 droop	25.0%	53.4%

Calculated 30 May 2026 using SusProg3D V5.160R (1741.17)

Figure 51 susprog3d rear suspension "roll bump" section results

subproject future 2027 no swearing about.ssd front steering

RHD Rack and pinion, rear steer

Vehicle horizontal datum (Y): Vehicle centreline
Vehicle vertical datum (Z): Ground
Vehicle longitudinal datum (X): Bulkhead

Steering pivot points (from vehicle Y, Z, X datum)
- tie rod (rack ball joint) - Y 310.00
- Z 183.56
- X -749.01
- tie rod (steering arm) - Y 657.07
- Z 215.33
- X -634.59

Steering tie rod length 366.82
Steering arm length (actual) 60.21
Steering arm length (effective) 55.32
Steering arm - tie rod angle (effective, degrees) 113.24

Rack c-factor 80.00
Steering ratio - nominal (straight ahead) 4.61
Turning radius (tyre contact centre) LH/RH 3125.40
Turning circle (curb-to-curb) LH/RH 6456.54
Steering wheel turns, lock to lock 0.82

Wheel toe reference length 500.00

Bump steer	absolute mm	relative mm
26.00 bump	-0.23 toe out	-0.01
25.00 bump	-0.23 toe out	-0.01
20.00 bump	-0.23 toe out	-0.01
15.00 bump	-0.22 toe out	0.00
10.00 bump	-0.22 toe out	0.00
5.00 bump	-0.22 toe out	0.00
Static	-0.22 toe out	0.00
5.00 droop	-0.22 toe out	0.00
10.00 droop	-0.22 toe out	0.00
15.00 droop	-0.22 toe out	0.00
20.00 droop	-0.22 toe out	0.00
25.00 droop	-0.22 toe out	0.00
26.00 droop	-0.22 toe out	0.00

Steering turn angle	Strg wheel rotation	Toe out in turn	Wheel toe angle LH RH	Rack travel	Steering Ratio:1 LH RH	Ackermann %
30 LH	147	-13.42	-43.42 30.00	32.77	2.24 5.01	70.3%
25 LH	122	-7.23	-32.23 25.00	27.20	3.05 5.02	55.6%
20 LH	97	-4.00	-24.00 20.00	21.63	3.59 4.99	50.1%
15 LH	72	-2.05	-17.05 15.00	16.08	3.99 4.93	48.5%
10 LH	48	-0.88	-10.88 10.00	10.61	4.29 4.83	49.9%
5 LH	24	-0.25	-5.25 5.00	5.24	4.52 4.70	60.4%
Straight	0	-0.05	-0.03 -0.03	0.00	4.61 4.61	0.0%
5 RH	24	-0.25	5.00 -5.25	5.24	4.70 4.52	60.4%
10 RH	48	-0.88	10.00 -10.88	10.61	4.83 4.29	49.9%
15 RH	72	-2.05	15.00 -17.05	16.08	4.93 3.99	48.5%
20 RH	97	-4.00	20.00 -24.00	21.63	4.99 3.59	50.1%
25 RH	122	-7.23	25.00 -32.23	27.20	5.02 3.05	55.6%
30 RH	147	-13.42	30.00 -43.42	32.77	5.01 2.24	70.3%

Steering turn angle	Camber (actual) LH RH	Camber (change) LH RH	Caster (actual) LH RH	Caster (change) LH RH
30 LH	3.44 -3.08	4.64 -1.88	6.68 2.14	1.68 -2.86
25 LH	2.14 -2.87	3.34 -1.67	6.59 2.67	1.59 -2.33
20 LH	1.20 -2.62	2.40 -1.42	6.37 3.19	1.37 -1.81
15 LH	0.45 -2.33	1.65 -1.13	6.08 3.69	1.08 -1.31
10 LH	-0.19 -1.99	1.01 -0.79	5.75 4.16	0.75 -0.84
5 LH	-0.73 -1.62	0.47 -0.42	5.39 4.60	0.39 -0.40
Straight	-1.20 -1.20	0.00 0.00	5.00 5.00	0.00 0.00
5 RH	-1.62 -0.73	-0.42 0.47	4.60 5.39	-0.40 0.39
10 RH	-1.99 -0.19	-0.79 1.01	4.16 5.75	-0.84 0.75
15 RH	-2.33 0.45	-1.13 1.65	3.69 6.08	-1.31 1.08
20 RH	-2.62 1.20	-1.42 2.40	3.19 6.37	-1.81 1.37
25 RH	-2.87 2.14	-1.67 3.34	2.67 6.59	-2.33 1.59
30 RH	-3.08 3.44	-1.88 4.64	2.14 6.68	-2.86 1.68

Steering turn angle	Jacking effect		Caster trail		KPI Offset		Steering offset	
	LH	RH	LH	RH	LH	RH	LH	RH
30 LH	0.71	-0.70	25.91	8.20	0.64	0.72	25.92	8.23
25 LH	0.66	-0.61	25.58	10.28	0.64	0.71	25.59	10.30
20 LH	0.55	-0.51	24.70	12.29	0.65	0.71	24.71	12.31
15 LH	0.42	-0.39	23.55	14.21	0.65	0.70	23.56	14.23
10 LH	0.28	-0.27	22.24	16.03	0.66	0.69	22.25	16.04
5 LH	0.14	-0.13	20.82	17.74	0.67	0.69	20.83	17.75
Straight	0.00	0.00	19.32	19.32	0.68	0.68	19.33	19.33
5 RH	-0.13	0.14	17.74	20.82	0.69	0.67	17.75	20.83
10 RH	-0.27	0.28	16.03	22.24	0.69	0.66	16.04	22.25
15 RH	-0.39	0.42	14.21	23.55	0.70	0.65	14.23	23.56
20 RH	-0.51	0.55	12.29	24.70	0.71	0.65	12.31	24.71
25 RH	-0.61	0.66	10.28	25.58	0.71	0.64	10.30	25.59
30 RH	-0.70	0.71	8.20	25.91	0.72	0.64	8.23	25.92

Datum reference dimensions
Chassis horizontal datum (Y): Chassis centreline
Chassis vertical datum (Z): Ground
Chassis longitudinal datum (X): Bulkhead

Calculated using SusProg3D V5.160R (1741.17) by DMURacing_Laptop 31/05/2026 20:32

Chassis pivot points (from chassis Y, Z, X datum)
- tie rod (rack ball joint) - Y 310.00
- Z 141.85
- X -750.00

Upright pivot points (from upright Y, Z, X datum)
Camber shim adjusts both top pivot and toe control pivot
- camber shim stack (included in Y dimensions) 10.00
- camber shim thickness (one shim) 2.00
- camber shim count 5
- tie rod (steering arm) - Y 40.00
- Z 0.00
- X -60.00

Upright pivot points (from chassis Y, Z, X datum)
- tie rod (steering arm) - Y 657.07
- Z 174.40
- X -635.80

Calculated 31 May 2026 using SusProg3D V5.160R (1741.17)

Figure 52 susprog3d front suspension "steering" section results

SusProg3D future 2027 no swearing about.s3d Rear Steering

Toe control link - chassis

Vehicle horizontal datum (Y): Vehicle centreline
Vehicle vertical datum (Z): Ground
Vehicle longitudinal datum (X): Bulkhead

Chassis pivot points (from vehicle Y, Z, X datum)

- toe control link	- Y	294.00
	- Z	168.86
	- X	-2195.06
- toe control link upright pivot	- Y	652.75
	- Z	170.45
	- X	-2195.11

Toe control link length 358.75

Wheel toe reference length 500.00

Bump steer	absolute mm	relative mm
31.00 bump	0.47 toe in	-0.01
30.00 bump	0.47 toe in	-0.01
25.00 bump	0.48 toe in	0.00
20.00 bump	0.48 toe in	0.00
15.00 bump	0.48 toe in	0.00
10.00 bump	0.48 toe in	0.00
5.00 bump	0.48 toe in	0.00
Static	0.48 toe in	0.00
5.00 droop	0.48 toe in	0.00
10.00 droop	0.48 toe in	0.00
15.00 droop	0.47 toe in	-0.01
20.00 droop	0.47 toe in	-0.01
25.00 droop	0.47 toe in	-0.01

Datum reference dimensions

Chassis horizontal datum (Y): Chassis centreline
Chassis vertical datum (Z): Ground
Chassis longitudinal datum (X): Bulkhead

Chassis pivot points (from chassis Y, Z, X datum)

- toe control link	- Y	294.00
	- Z	117.25
	- X	-2195.91

Upright pivot points (from upright Y, Z, X datum)

- toe control link	- Y	40.00
	- Z	-50.00
	- X	-70.00

Upright pivot points (from chassis Y, Z, X datum)

- toe control link	- Y	652.75
	- Z	118.84
	- X	-2195.98

Calculated 30 May 2026 using SusProg3D V5.160R (1741.17)

Figure 53 susprog3d rear suspension "steering" results

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susprog3d future 2021 no swearing about.s3d front suspension

Vehicle horizontal datum (Y): Vehicle centreline
Vehicle vertical datum (Z): Ground
Vehicle longitudinal datum (X): Bulkhead

LH and RH side identical

Wheel rate      Motion      Length      Spring      Arb
N/mm      lb/in      ratio      deflection      twist

Shockabsorber      11.68      66.70      1.936      201.93      24.03
Oil spring

Pushrod and bellcrank
Pushrod mounting (A-arm)      - Y      578.33
                              - Z      154.29
                              - X      -571.21
Pushrod length      380.62
Bellcrank pivots      - Y      282.18      325.00      400.15      327.49
                              - Z      488.87      439.99      505.91      440.50
                              - X      -498.71      -497.25      -494.50      -577.21
Pushrod arm length_offset      80.00      0.00
Shockabsorber arm length_offset      65.00      0.00
Pushrod pivot to shockabsorber pivot      -102.74
Angle between arms      -89.61
Clockwise rotation in bump (LH)

Shockabsorber
Shockabsorber mounting (chassis)      - Y      275.00
                              - Z      488.62
                              - X      -296.91
Length - compressed (full bump)      0.00
      static      201.93
      extended (full droop)      0.00
Stroke      0.00
Motion ratio (static)      1.94

Corner weight (unsprung)      16.010 kg      35.30 lb
Corner weight (sprung)      55.410 kg      122.16 lb
Corner weight (total)      71.420 kg      157.45 lb
Suspension frequency      2.31 Hz      138.65 cpm

Oil spring
Spring seat (shockabsorber shaft)      36.20 mm      1.425 in
Spring seat (shockabsorber body)      67.76 mm      2.668 in
Spring rate      43.78 N/mm      250.00 lb/in
Static load      1051.98 N      107.272 kg      236.49 lb
Static length      97.97 mm      3.857 in
Free length      122.00 mm      4.803 in
Compressed length      84.39 mm      3.323 in
Preload      10.73 mm      0.422 in

Oil spring parameters
Spring rate      43.78 N/mm      250.00 lb/in
Type of ends      Closed and ground
Total number of coils      7.00
Number of active coils      5.00
Coil ID      36.30 mm      1.429 in
Wire dia      6.44 mm      0.254 in
Free length      122.00 mm      4.803 in
Solid length      41.89 mm      1.649 in
Solid load      3507.47 N      357.662 kg      788.51 lb
Natural frequency      252.54 Hz
Torsional stress (corrected) - static      524.49 MPa      76070.34 psi
                              - full bump      820.89 MPa      119060.38 psi
                              - solid      1748.72 MPa      253630.46 psi

Chassis pivot points (from chassis Y, Z, X datum)
- shockabsorber mounting      - Y      275.00
                              - Z      450.00
                              - X      -300.00
- bellcrank pivot (P1)      - Y      325.00
                              - Z      400.00
                              - X      -500.00
- bellcrank pivot axis (P2 offset from P1)      - Y      75.15
                              - Z      65.94
                              - X      2.30
A-arm pivot points (Y from A-arm chassis pivot, Z from A-arm plane, X from A-arm normal)
- shockabsorber mounting      - Y      120.00
                              - Z      -60.00
                              - X      15.00
- coil spring mounting      - Y      120.00
                              - Z      -60.00
                              - X      15.00
A-arm pivot points (Y from A-arm apex, Z from A-arm plane, X from A-arm normal)
- Pushrod mounting      - Y      90.00
                              - Z      25.00
                              - X      -5.00

Calculated 31 May 2026 using SusProg3D V5.160R (1741.17)

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Figure 54 susprog3d front suspension "spring, shock & ARB" section results

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SusProg3D future 2027 no swearing about.s3d Rear Suspension

/ehicle horizontal datum (Y): Vehicle centreline
/ehicle vertical datum (Z): Ground
/ehicle longitudinal datum (X): Bulkhead

LH and RH side identical

Wheel rate      Motion      Length      Spring      Arb
N/mm      lb/in      ratio      deflection      twist

Shockabsorber      2.191      175.93
Coil spring      9.12      52.08      2.191      93.70      28.30

Pushrod and bellcrank
- Pushrod mounting (A-arm)      - Y      548.61
      - Z      154.64
      - X      -2093.74
- Pushrod length      269.26
- bellcrank pivots      - Y      366.60      P1 Bellcrank P2      Pushrod
      - Z      421.91      375.00      347.23      445.48
      - X      -2046.33      -2047.76      -2143.81      -2068.75
- Pushrod arm length _offset      80.00      0.00
- shockabsorber arm length _offset      52.00      0.00
- Pushrod pivot to shockabsorber pivot      84.36
- angle between arms      76.18
- anti-clockwise rotation in bump (LH)

Shockabsorber
- shockabsorber mounting (chassis)      - Y      200.00
      - Z      393.27
      - X      -1997.60
- length - compressed (full bump)      143.00
- static      175.93
- extended (full droop)      200.00
- stroke      57.00
- motion ratio (static)      2.19

Corner weight (unsprung)      16.670 kg      36.75 lb
Corner weight (sprung)      57.670 kg      127.14 lb
Corner weight (total)      74.340 kg      163.89 lb
Suspension frequency      2.00 Hz      120.09 cpm

Coil spring
- spring seat (shockabsorber shaft)      30.00 mm      1.181 in
- spring seat (shockabsorber body)      52.23 mm      2.056 in
- spring rate      43.78 N/mm      250.00 lb/in
- static load      1239.10 N      126.353 kg      278.56 lb
- static length      93.70 mm      3.689 in
- free length      122.00 mm      4.803 in
- compressed length      79.55 mm      3.132 in
- preload      16.96 mm      0.668 in

Coil spring parameters
- spring rate      43.78 N/mm      250.00 lb/in
- type of ends      Closed and ground
- total number of coils      7.00
- number of active coils      5.00
- coil ID      36.30 mm      1.429 in
- wire dia      6.44 mm      0.254 in
- free length      122.00 mm      4.803 in
- solid length      41.89 mm      1.649 in
- solid load      3507.44 N      357.659 kg      788.50 lb
- natural frequency      252.54 Hz
- torsional stress (corrected) - static      617.78 MPa      89600.99 psi
      - full bump      926.58 MPa      134389.95 psi
      - solid      1748.70 MPa      253628.09 psi

Chassis pivot points (from chassis Y, Z, X datum)
- shockabsorber mounting      - Y      200.00
      - Z      343.00
      - X      -2000.00
- bellcrank pivot (P1)      - Y      375.00
      - Z      320.00
      - X      -2050.00
- bellcrank pivot axis (P2 offset from P1)      - Y      -27.77
      - Z      -2.53
      - X      -96.03
A-arm pivot points (Y from A-arm chassis pivot, Z from A-arm plane, X from A-arm normal)
- shockabsorber mounting      - Y      130.00
      - Z      -60.00
      - X      40.00
- coil spring mounting      - Y      130.00
      - Z      -60.00
      - X      40.00
A-arm pivot points (Y from A-arm apex, Z from A-arm plane, X from A-arm normal)
- Pushrod mounting      - Y      80.00
      - Z      18.00
      - X      30.00

Calculated 30 May 2026 using SusProg3D V5.160R (1741.17)

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Figure 55 susprog3d rear suspension "spring, shock & ARB" section results

SusProg3D future 2027 no swearing about.s3d Dynamic

Tyre rolling radius (nominal)	front/rear	220.98 mm	220.98 mm
Track	front/rear	1359.22 mm	1350.24 mm
Spring base	front/rear	1359.22 mm	1350.24 mm
Wheelbase		1550.26 mm	
Vehicle CofG	- from ground	322.30 mm	
	- from front axle cl	790.64 mm	
Sprung CofG	- from ground	351.58 mm	
	- from front axle cl	792.72 mm	
Wheel rates in bump			
- front	- tyre	112.96 N/mm	645.02 lb/in
	- spring	11.68 N/mm	66.71 lb/in
	- combined	10.59 N/mm	60.46 lb/in
- rear	- tyre	42.70 N/mm	243.82 lb/in
	- spring	9.12 N/mm	52.08 lb/in
	- combined	7.52 N/mm	42.91 lb/in
Wheel rates in roll			
- front	- tyre	112.96 N/mm	645.02 lb/in
	- spring	11.68 N/mm	66.71 lb/in
	- combined	10.59 N/mm	60.46 lb/in
- rear	- tyre	42.70 N/mm	243.82 lb/in
	- spring	9.12 N/mm	52.08 lb/in
	- combined	7.52 N/mm	42.91 lb/in

Frequency (in bump)			
- sprung mass	- front	2.20 Hz	131.83 cpm
- unsprung mass	- front	14.06 Hz	843.67 cpm
- sprung mass	- rear	1.81 Hz	108.51 cpm
- unsprung mass	- rear	8.92 Hz	534.90 cpm

Roll stiffness - front	- tyres	104345 N·m/rad	76961 lb-ft/rad	90.6%
	- spring	10792 N·m/rad	7960 lb-ft/rad	9.4%
	- combined	9780 N·m/rad	7214 lb-ft/rad	
Roll stiffness - rear	- tyres	38924 N·m/rad	28709 lb-ft/rad	82.4%
	- spring	8314 N·m/rad	6132 lb-ft/rad	17.6%
	- combined	6851 N·m/rad	5053 lb-ft/rad	
Roll stiffness - car	- tyres	143270 N·m/rad	105670 lb-ft/rad	88.2%
	- spring	19106 N·m/rad	14092 lb-ft/rad	11.8%
	- combined	16631 N·m/rad	12266 lb-ft/rad	
Roll stiffness distribution - front		58.8%		

Roll direction LH (RH turn)

Test #	Lateral Accel G	Roll centre height		Sprung CoG above roll axis	Weight transfer		LH	Corner weights			Rear RH	Front	Tyr
		Front	Rear		Front	Rear		Front	RH	LH			
1	1.00	49.98	61.79	295.56	39.70	30.91	111.12	31.72	105.25		43.43		0.29

Calculated 30 May 2026 using SusProg3D V5.160R (1741.17)

Figure 56 susprog3d suspension dynamics results

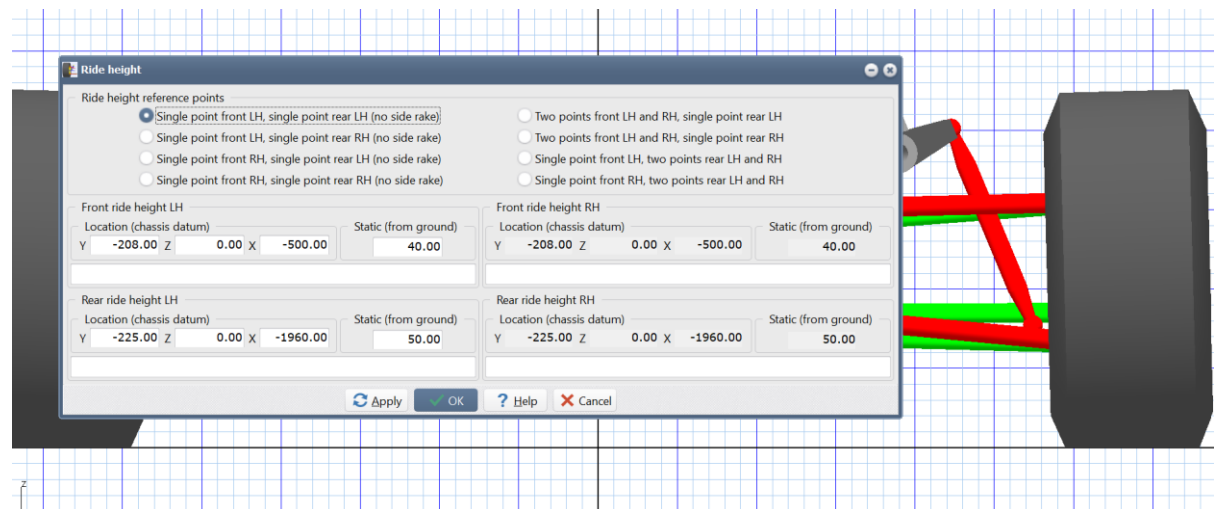


Figure 57 chassis datum and ride height section of susprog3d

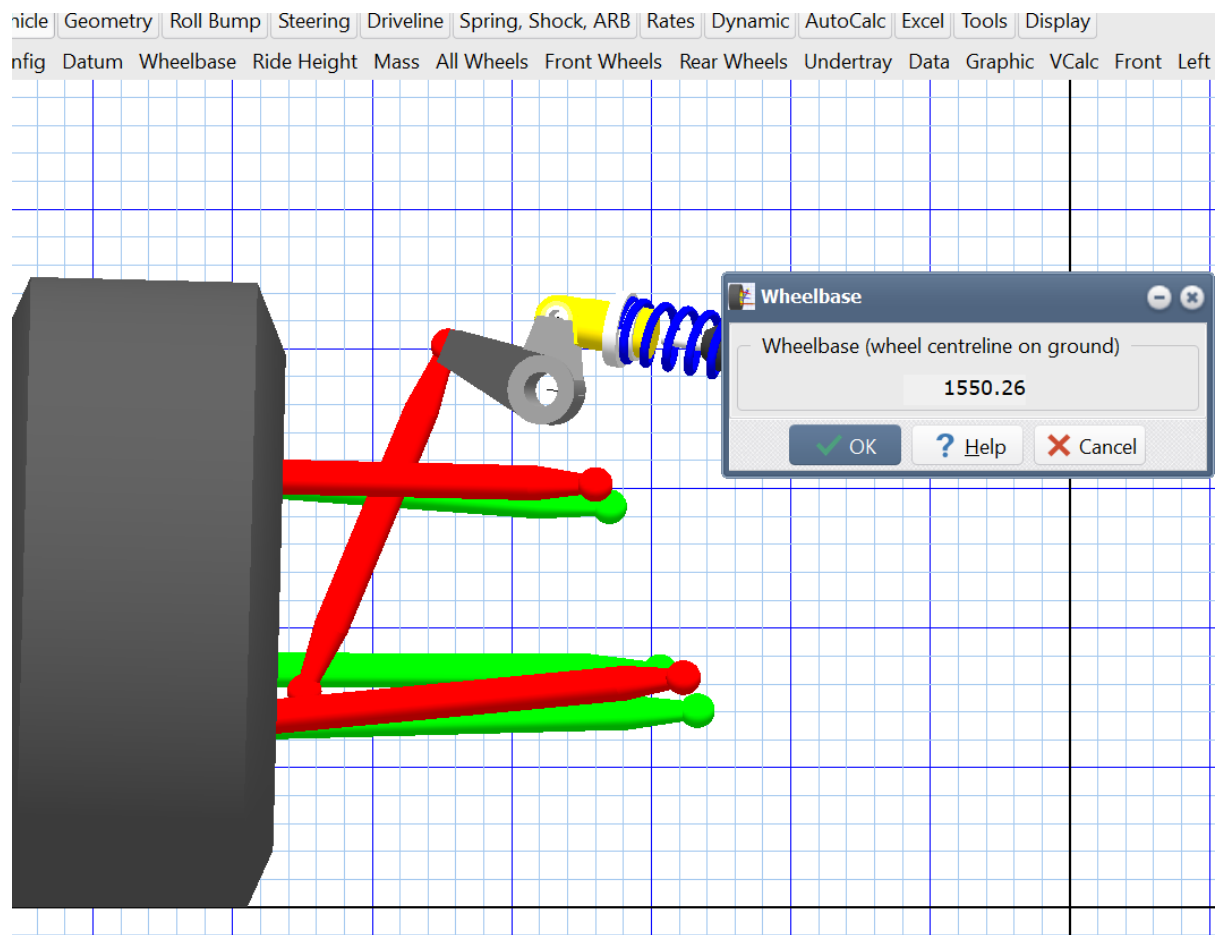


Figure 58 wheelbase length in mm according to finalised susprog3d geometry